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United States Patent	12406692
Kind Code	B2
Date of Patent	September 02, 2025
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Magnetic tape, magnetic tape cartridge, and magnetic tape apparatus

Abstract

The magnetic tape includes a non-magnetic support, and a magnetic layer containing a ferromagnetic powder. A magnetic tape cartridge and a magnetic tape apparatus include the magnetic tape. A core void volume V_{vc} specified in international organization for standardization (ISO) 25178 on a surface of the magnetic layer is 1.200 nm.sup.3 or more and 3.000 nm.sup.3 or less, and standard deviation σ of the V_{vc} in a width direction of the surface of the magnetic layer is 0.120 nm.sup.3 or less.

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Appl. No.: 18/758432

Filed: June 28, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250014599 A1	Jan. 09, 2025

Foreign Application Priority Data

JP	2023-107623	Jun. 30, 2023
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Publication Classification

Int. Cl.: G11B5/73 (20060101); G11B5/588 (20060101)

U.S. Cl.:

CPC G11B5/736 (20190501); G11B5/588 (20130101);

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority under 35 U.S.C 119 to Japanese Patent Application No. 2023-107623 filed on Jun. 30, 2023. The above application is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a magnetic tape, a magnetic tape cartridge, and a magnetic tape

apparatus.

2. Description of the Related Art

(3) There are two types of magnetic recording media: a tape shape and a disk shape, and a tape-shaped magnetic recording medium, that is, a magnetic tape is mainly used for data storage applications such as data backup and archiving (for example, see JP2016-524774A, US2019/0164573A1, and JP6590102B).

SUMMARY OF THE INVENTION

(4) Recording of data on a magnetic tape is usually performed by running the magnetic tape in a magnetic tape apparatus (generally called a “drive”) and recording the data on a data band by making a magnetic head follow the data band of the magnetic tape. Thereby, a data track is formed in the data band. In addition, in a case where the recorded data is reproduced, the data recorded on the data band is read by running the magnetic tape in the magnetic tape apparatus and by making the magnetic head follow the data band of the magnetic tape. After such recording or reproduction, the magnetic tape is usually stored in a state of being wound around a reel inside a magnetic tape cartridge until the next recording and/or reproduction is performed.

(5) In order to increase an accuracy of the magnetic head following the data band of the magnetic tape in recording and/or reproduction as described above, a system for performing head tracking using a servo signal (hereinafter, it is described as a “servo system”) has been put into practical use. However, in a case where recording and/or reproduction is performed after the above-mentioned storage, tape width deformation of the magnetic tape caused by the storage may cause a phenomenon (generally called “off-track”) in which the magnetic head for recording and/or reproducing the data deviates from a target track position. In a case where overwriting of the recorded data, reproduction failure, or the like occurs due to off-track, operational stability of the drive may deteriorate. In recent years, off-track is more likely to occur due to an increase in track density associated with an increase in capacity of the magnetic tape, so that an increasing need to improve the operational stability of the drive has been made. On the other hand, in recent years, in the field of data storage, long-term storage of data, which is called archive, has been performed. However, in general, the longer the storage period, the more likely the tape width deformation of the magnetic tape is to occur, and an occurrence frequency of the off-track tends to increase.

(6) In view of the above, an object of an aspect of the present invention is to provide a magnetic tape that can contribute to improvement of operational stability of a drive in recording and/or reproduction after long-term storage.

(7) One aspect of the present invention is as follows.

(8) [1] A magnetic tape comprising: a non-magnetic support; and a magnetic layer containing a ferromagnetic powder, in which a core void volume V_{vc} (hereinafter, also referred to as “ V_{vc} of a magnetic layer surface” or simply “ V_{vc} ”) specified in international organization for standardization (ISO) 25178 on a surface of the magnetic layer is 1.200 nm.sup.3 or more and 3.000 nm.sup.3 or less, and standard deviation σ of the V_{vc} in a width direction of the surface of the magnetic layer (hereinafter, also referred to as a “width direction σ of V_{vc} ”) is 0.120 nm.sup.3 or less.

(9) [2] The magnetic tape according to [1], in which the standard deviation σ of the V_{vc} is 0.095 nm.sup.3 or less.

(10) [3] The magnetic tape according to [1] or [2], further comprising: a non-magnetic layer containing a non-magnetic powder between the non-magnetic support and the magnetic layer.

(11) [4] The magnetic tape according to [3], in which the non-magnetic powder of the non-magnetic layer includes an Fe-based inorganic oxide powder having an average particle volume of 2.0×10^{-6} μm^3 or less.

(12) [5] The magnetic tape according to [3] or [4], in which the non-magnetic powder of the non-magnetic layer includes carbon black having a pH of 9.0 or less.

(13) [6] The magnetic tape according to any one of [3] to [5], in which a thickness of the non-magnetic layer is 0.1 μm or more and 0.7 μm or less.

- (14) [7] The magnetic tape according to any one of [1] to [6], further comprising: a back coating layer containing a non-magnetic powder on a surface side of the non-magnetic support opposite to a surface side having the magnetic layer.
- (15) [8] The magnetic tape according to [7], in which the non-magnetic powder of the back coating layer includes a non-magnetic powder having an average plate diameter of 50 nm or more and 1,000 nm or less and an average plate thickness of 12 nm or less.
- (16) [9] The magnetic tape according to any one of [1] to [8], in which a tape thickness is 5.2 μm or less.
- (17) [10] The magnetic tape according to any one of [1] to [9], in which a tape thickness is 5.0 μm or less.
- (18) [11] The magnetic tape according to any one of [1] to [10], in which a vertical squareness ratio of the magnetic tape is 0.60 or more.
- (19) [12] The magnetic tape according to any one of [1] to [11], in which a vertical squareness ratio of the magnetic tape is 0.65 or more.
- (20) [13] The magnetic tape according to any one of [1] to [12], in which the standard deviation σ of the Vvc is 0.095 nm.sup.3 or less, the magnetic tape further includes a non-magnetic layer containing a non-magnetic powder between the non-magnetic support and the magnetic layer, the non-magnetic powder of the non-magnetic layer includes an Fe-based inorganic oxide powder having an average particle volume of $2.0 \times 10^{-6} \mu\text{m}^3$ or less and carbon black having a pH of 9.0 or less, a thickness of the non-magnetic layer is 0.1 μm or more and 0.7 μm or less, the magnetic tape further includes a back coating layer containing a non-magnetic powder on a surface side of the non-magnetic support opposite to a surface side having the magnetic layer, the non-magnetic powder of the back coating layer includes a non-magnetic powder having an average plate diameter of 50 nm or more and 1,000 nm or less and an average plate thickness of 12 nm or less, a tape thickness is 5.0 μm or less, and a vertical squareness ratio of the magnetic tape is 0.65 or more.
- (21) [14] A magnetic tape cartridge comprising: the magnetic tape according to any one of [1] to [13].
- (22) [15] A magnetic tape apparatus comprising: the magnetic tape according to any one of [1] to [13].
- (23) [16] The magnetic tape apparatus according to [15], further comprising: a magnetic head, in which the magnetic head has a module including an element array with a plurality of magnetic head elements between a pair of servo signal reading elements, and the magnetic tape apparatus changes an angle θ formed by an axis of the element array with respect to a width direction of the magnetic tape during running of the magnetic tape in the magnetic tape apparatus.
- (24) According to one aspect of the present invention, it is possible to provide a magnetic tape that can contribute to improvement of operational stability of a drive in recording and/or reproduction after long-term storage. In addition, according to one aspect of the present invention, it is possible to provide a magnetic tape cartridge and a magnetic tape apparatus which include the magnetic tape.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows an example of a track profile in which plotting is performed, where a horizontal axis represents a track position and a vertical axis represents an output of a reproduction signal.
- (2) FIG. 2 shows an example of a graph relating to a non-linear component at an initial stage.
- (3) FIG. 3 shows an example of a graph relating to a non-linear component after storage.
- (4) FIG. 4 is an example of a graph showing an absolute value of a difference between a non-linear

component at an initial stage for each reproducing element and a non-linear component after storage (difference between before and after storage of a non-linear component), which are obtained each reproducing element.

(5) FIG. 5 is a schematic view showing an example of a module of a magnetic head.

(6) FIG. 6 is an explanatory diagram of a relative positional relationship between a module and a magnetic tape during magnetic tape running in a magnetic tape apparatus.

(7) FIG. 7 is an explanatory diagram relating to a change in an angle θ during magnetic tape running.

(8) FIG. 8 shows an example (schematic process diagram) of a manufacturing step of the magnetic tape.

(9) FIG. 9 shows an arrangement example of data bands and servo bands.

(10) FIG. 10 shows an arrangement example of a servo pattern of a linear tape-open (LTO) Ultrium format tape.

(11) FIG. 11 is an explanatory diagram of a measuring method of an angle θ during magnetic tape running.

(12) FIG. 12 is a schematic view showing an example of the magnetic tape apparatus.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) Magnetic Tape

(14) One aspect of the present invention relates to a magnetic tape including a non-magnetic support and a magnetic layer containing a ferromagnetic powder. In the magnetic tape, a core void volume V_{vc} specified in ISO 25178 on a surface of the magnetic layer is 1.200 nm.sup.3 or more and 3.000 nm.sup.3 or less, and standard deviation σ of the V_{vc} in a width direction of the surface of the magnetic layer is 0.120 nm.sup.3 or less.

(15) As a result of extensive studies, the present inventor has newly found that a magnetic tape in which the V_{vc} of the magnetic layer surface is in the above range and the width direction σ of the V_{vc} is in the above range can contribute to improvement of operational stability of a drive. Hereinafter, supposition of the present inventor regarding this point is described. Note that the present invention is not limited to the supposition described in the present specification.

(16) As described above, the tape width deformation generated by the long-term storage may cause a decrease in operational stability of the magnetic tape in the drive. In this regard, in recent years, it has been proposed to acquire information on dimensions in a width direction of the magnetic tape during running by using a servo signal and to change an angle (hereinafter, also referred to as a "head tilt angle") at which an axial direction of a module of a magnetic head is tilted against the width direction of the magnetic tape according to the acquired dimension information (see JP2016-524774A and US2019/0164573A1, for example, paragraphs 0059 to 0067 and 0084 of JP2016-524774A). In addition, the information on the dimensions in the width direction of the magnetic tape during running is acquired using the servo signal, and a tension applied in a longitudinal direction of the magnetic tape is adjusted according to the acquired dimension information, thereby controlling the dimensions in the width direction of the magnetic tape (see, for example, a paragraph 0171 of JP6590102B). For example, control means for a dynamic track position during running of the magnetic tape as described above can be means for suppressing off-track.

(17) However, the present inventor has made extensive studies in order to further improve the operational stability of the drive in recording and/or reproduction after long-term storage, and paid attention to the fact that there may be off-track factors for which it is difficult to compensate by the control means for the dynamic track position. Hereinafter, this point will be further described.

(18) In a case where the dynamic track position is controlled by changing the head tilt angle, a pitch of a magnetic head element (specifically, a recording element and/or a reproducing element) changes evenly according to the head tilt angle regardless of the position in the tape width direction. In a case where the dynamic track position is controlled by adjusting the tension applied in the longitudinal direction of the magnetic tape, the tension in the entire width of the tape is

usually adjusted, so that the tape width changes evenly by the tension adjustment regardless of the position in the tape width direction. In a case where a degree of the tape width deformation is homogeneous over the entire magnetic tape, that is, in a case where a tape width deformation component is only a linear component, it is possible to completely compensate for the off-track by the control means. Therefore, it is possible to completely registrate the data track and the magnetic head element. On the other hand, in a case where the degree of the tape width deformation varies depending on the position and is non-homogeneous, that is, in a case where the tape width deformation component includes a non-linear component, it is difficult to compensate for the off-track caused by the non-linear component, by the control means. The present inventor has considered that reduction in this non-linear component can contribute to suppression of the decrease in operational stability of the drive due to the off-track factor for which it is difficult to compensate by the control means for the dynamic track position. Regarding this point, the present inventor has considered that, in the magnetic tape, the V_{vc} of the magnetic layer surface being in the above range and the standard deviation σ of the V_{vc} being in the above range can contribute to suppression of occurrence of the non-linear component of the tape width deformation. Accordingly, the present inventor supposes that the magnetic tape can contribute to the improvement of the operational stability of the drive in the recording and/or the reproduction after the long-term storage.

(19) V_{vc} of Magnetic Layer Surface and Width Direction σ of V_{vc}

(20) In the present invention and the present specification, V_{vc} is a core void volume V_{vc} specified in ISO 25178. ISO 25178 is an ISO standard related to three-dimensional surface property parameters, and more specifically, ISO 25178-2:2021 (Geometrical product specifications (GPS)-Surface texture: Areal-Part 2: Terms, definitions and surface texture parameters).

(21) In order to obtain V_{vc} of the magnetic layer surface, as described below, the measurement is performed on the surface of the magnetic layer of the magnetic tape using an atomic force microscope (AFM). In the present invention and the present specification, the term “magnetic layer surface (surface of the magnetic layer)” has the same meaning as the surface of the magnetic tape on the magnetic layer side.

(22) The measurement region is a region of 40 μm square (40 μm ×40 μm). The measurement is executed at five points. The five points where measurement on the magnetic layer surface is performed are set to five points where longitudinal direction positions are the same but the width direction positions are different. The longitudinal direction position is randomly selected on the magnetic layer surface, the width direction is divided into five parts with respect to the magnetic tape width at this longitudinal direction position (therefore, in a case where the magnetic tape width is defined as W , the width of each division is “ $W/5$ ”), and V_{vc} specified in ISO 25178 is obtained for a region of 40 μm square (40 μm ×40 μm) randomly selected in each division. V_{vc} can be calculated using known AFM data analysis software. Examples of the AFM data analysis software include AFM data analysis software (Nanoscope Analysis) provided by BRUKER. An arithmetic average of the five measured values thus obtained is defined as the V_{vc} of the magnetic layer surface of the magnetic tape to be measured. In addition, standard deviation σ (that is, positive square root of the dispersion) of the five measured values thus obtained is defined as standard deviation σ of the V_{vc} (width direction σ of V_{vc}) in the width direction of the surface of the magnetic layer of the magnetic tape to be measured.

(23) The following measurement conditions can be used as an example of the AFM measurement conditions.

(24) The region of the area of 40 μm ×40 μm on the surface of the magnetic layer of the magnetic tape is measured with an AFM (Nanoscope 5 manufactured by BRUKER) in a peak fore-tapping mode. SCANASYST-AIR manufactured by BRUKER is used as a probe, a resolution is set to 512 pixels×512 pixels, and a scan speed is set to a speed at which one screen (512 pixels×512 pixels) is measured in 512 seconds.

(25) Vvc of Magnetic Layer Surface

(26) The Vvc of the magnetic layer surface of the magnetic tape is 1.200 nm.³ or more and 3.000 nm.³ or less.

(27) The Vvc of the magnetic layer surface is 1.200 nm.³ or more, preferably 1.500 nm.³ or more, and more preferably 1.800 nm.³ or more from the viewpoint of improving the operational stability of the drive in recording and/or reproduction after long-term storage. In addition, from the above viewpoint, the Vvc of the magnetic layer surface is 3.000 nm.³ or less, preferably 2.800 nm.³ or less, more preferably 2.500 nm.³ or less, still more preferably 2.300 nm.³ or less, and still more preferably 2.000 nm.³ or less.

(28) Width Direction σ of Vvc

(29) The standard deviation σ of the Vvc in the width direction of the surface of the magnetic layer of the magnetic tape (width direction σ of Vvc) is 0.120 nm.³ or less, preferably 0.100 nm.³ or less, more preferably 0.095 nm.³ or less, still more preferably 0.090 nm.³ or less, still more preferably 0.085 nm.³ or less, and still more preferably 0.080 nm.³ or less from the viewpoint of improving the operational stability of the drive in recording and/or reproduction after long-term storage. The width direction σ of Vvc may be, for example, 0.000 nm.³ or more, 0.001 nm.³ or more, 0.010 nm.³ or more, 0.020 nm.³ or more, 0.030 nm.³ or more, 0.040 nm.³ or more, 0.050 nm.³ or more, or 0.060 nm.³ or more. From the viewpoint of improving the operational stability of the drive in recording and/or reproduction after long-term storage, it is supposed that the smaller the value of the width direction σ of the Vvc is, the more preferable it is.

(30) A control method of the Vvc of the magnetic layer surface and a control method of the width direction σ of the Vvc will be described below.

(31) Non-Linear Component of Tape Width Deformation

(32) The present inventor considers that “the non-linear component in the tape width direction generated by storage for 10 days in an environment of a temperature of 60° C. and a relative humidity of 20%” obtained by the following method can be an index of the non-linear component of the tape width deformation described above. A storage condition of “storage for 10 days in an environment of a temperature of 60° C. and a relative humidity of 20%” is employed as an example of storage conditions in an accelerated environment corresponding to long-term storage of data called archive, and the magnetic tape is not limited to a magnetic tape that is stored under such storage conditions.

(33) The following operation and measurement are performed in an environment of a temperature of 20° C. to 25° C. and a relative humidity of 40% to 60%, unless otherwise noted.

(34) The magnetic tape to be measured is a magnetic tape having a length of 200 μ m or more.

(35) By using a device having a winding mechanism for winding the magnetic tape by applying a tension in the longitudinal direction of the magnetic tape, the magnetic tape to be measured is wound around a magnetic tape reel having a hub diameter (outer diameter, the same applies hereinafter) of 44 mm, with a tension of 0.6 N (Newton) applied in the longitudinal direction of the magnetic tape. The magnetic tape wound around the reel in this manner is stored for 24 hours or longer in an environment where a temperature is 20° C. to 25° C. and a relative humidity is 40% to 60%, before the following measurement.

(36) Regarding one end part and the other end part of the magnetic tape, an end part on a start point side of winding to the reel is referred to as a tape inner peripheral end part, and the other end part is referred to as a tape outer peripheral end part. The following measurement is executed in a central wrap of each data band in a region (hereinafter, referred to as “tape outer peripheral region”) within 100 m from the tape outer peripheral end part and a region (hereinafter, referred to as “tape inner peripheral region”) within 100 m from the tape inner peripheral end part.

(37) The following measurement is performed using a magnetic head comprising a reproducing module including an element array with 10 channels or more of a reproducing element having an

element width (specifically, a reproducing element width) of $0.2\text{ }\mu\text{m}$ or more and $1.0\text{ }\mu\text{m}$ or less between a pair of servo signal reading elements, and a recording module including an element array with 10 channels or more of a recording element having an element width (specifically, a recording element width) of $1.2\text{ }\mu\text{m}$ or more and $2.9\text{ }\mu\text{m}$ or less between a pair of servo signal reading elements. The term “element width” refers to a physical dimension of the element width, and can be measured by an optical microscope, a scanning electron microscope, or the like. In the recording module, an interval between two adjacent recording elements in the head width direction is $83.25\text{ }\mu\text{m}$. In the reproducing module, an interval between two adjacent reproducing elements in the head width direction is $83.25\text{ }\mu\text{m}$. The above-mentioned interval is an interval between central portions of two adjacent recording elements in the recording module, is an interval between central portions of two adjacent reproducing elements in the reproducing module, and can be measured by an optical microscope or the like. In the measurement of Examples and Comparative Examples described below, a magnetic head comprising a recording module including an element array with 32 channels (0-channel to 31-channel) of a recording element between a pair of servo signal reading elements and a reproducing module including an element array with 32 channels (0-channel to 31-channel) of a reproducing element between a pair of servo signal reading elements was used.

(38) The reel around which the magnetic tape to be measured is wound and the magnetic head are mounted to a tape transport system of the magnetic tape apparatus to record and reproduce data. The tape transport system is mounted to a recording and reproducing amplifier capable of driving the magnetic head element (specifically, a recording element and a reproducing element) of the magnetic head. The recording and reproducing amplifier can be controlled from a personal computer (PC) via a controller. The magnetic head is mounted on an actuator (piezo motor or voice coil motor (VCM)) that operates in the tape width direction, and can be servo-followed such that the magnetic head is located at a certain track position during tape running based on a servo signal of the magnetic tape. In addition, in order to compensate for the linear component of the tape width deformation, the dynamic track position can be controlled by changing the head tilt angle of the magnetic head such that a difference between signals (PES1, PES2) of a reading position error signal (PES) in the width direction based on the servo signals obtained by two upper and lower servo signal reading elements is constant. During the following recording and reproduction, the servo-following and the dynamic track position control are executed.

(39) Next, a direct current (DC) pattern is recorded on a first wrap, a single frequency signal of 255 kfc/i is recorded on a second wrap, and the DC pattern is recorded on a third wrap for three continuous wraps in which the running direction is the same while running the magnetic tape at a constant speed of 3.0 m/sec . The unit “kfc/i” is a unit of a linear recording density (cannot be converted into an SI unit system).

(40) Single (shingled) recording of three or more tracks is performed such that a difference between $(\text{PES1} + \text{PES2})/2$ is 1200 nm . Single recording is also called shingled magnetic recording.

(41) Next, data is reproduced over a length of $90\text{ }\mu\text{m}$ in a region (tape outer peripheral region) within $100\text{ }\mu\text{m}$ from the tape outer peripheral end part and a region (tape inner peripheral region) within $100\text{ }\mu\text{m}$ from the tape inner peripheral end part for the center wraps of the three continuous wraps. The reproduction signal waveform and the servo signal waveform are acquired and saved using an oscilloscope. In each measurement, the track position of the reproducing element is moved at intervals of $1/30$ or less of the track pitch with respect to the tape width direction. The “output of the reproduction signal” is calculated for each reproducing element from the reproduction signal waveform acquired and saved using the above-mentioned oscilloscope, and the track position is calculated from the servo signal waveform. From these results, a track profile is created by performing plotting in which a horizontal axis represents the track position and a vertical axis represents the output of the reproduction signal. FIG. 1 shows an example of the track profile thus created.

(42) A median value between two track positions reduced by 1 dB or more from the maximum value of the output of the reproduction signal is obtained, the median value is plotted on the vertical axis for each reproducing element, and a linear approximation line is obtained by linear fitting using a least square method. For each of the tape outer peripheral region and the tape inner peripheral region, a difference between the linear approximation line and the actually measured value is obtained for each reproducing element, and the obtained difference is referred to as a “non-linear component at initial stage”. FIG. 2 shows an example of a graph relating to the non-linear component at an initial stage. In the example shown in FIG. 2 and the example shown in FIG. 3 described below, the number of the reproducing elements (number of channels) is 32 channels (reproducing element number (No.): 0-channel to 31-channel).

(43) After the measurement of the non-linear component at the initial stage, by using a device having a winding mechanism for winding the magnetic tape by applying a tension in the longitudinal direction of the magnetic tape, the magnetic tape to be measured is wound around a magnetic tape reel having a hub diameter of 44 mm, with a tension of 0.6 N applied in the longitudinal direction of the magnetic tape. In a case of such winding, winding is performed such that an end part that has served as the tape inner peripheral end part in winding around the reel before the measurement of the non-linear component at the initial stage becomes a tape inner peripheral end part. The magnetic tape wound around the reel in this manner is stored for 10 days in an environment of a temperature of 60° C. and a relative humidity of 20%.

(44) After the above-mentioned storage, the magnetic tape to be measured is stored in an environment of a temperature of 20° C. to 25° C. and a relative humidity of 40% to 60% for 24 hours or longer (where, up to 120 hours) in a state of being wound around the reel, and then the entire length of the magnetic tape to be measured is reciprocated once (forward and reverse) using the magnetic tape apparatus used for the measurement of the non-linear component at an initial stage. Data is reproduced over a length of 90 m in a region (tape outer peripheral region) within 100 m from the tape outer peripheral end part and a region (tape inner peripheral region) within 100 m from the tape inner peripheral end part under the same recording element, reproducing element, magnetic tape apparatus, and reproduction conditions as in the above-described measurement of the non-linear component at an initial stage. The output of the reproduction signal and the servo signal waveform are acquired and saved using the same oscilloscope as in the measurement of the non-linear component at an initial stage. The track position of the reproducing element is moved in the tape width direction for each measurement at the same interval as in the above-described measurement of the non-linear component at an initial stage. The “output of the reproduction signal” is calculated for each reproducing element from the reproduction signal waveform acquired and saved using the above-mentioned oscilloscope, and the track position is calculated from the servo signal waveform. From these results, a track profile is created by performing plotting in which a horizontal axis represents the track position and a vertical axis represents the output of the reproduction signal. FIG. 1 is also an example of the track profile thus created.

(45) A median value between two track positions reduced by 1 dB or more from the maximum value of the output of the reproduction signal is obtained, the median value is plotted on the vertical axis for each reproducing element, and a linear approximation line is obtained by linear fitting using a least square method. For each of the tape outer peripheral region and the tape inner peripheral region, a difference between the linear approximation line and the actually measured value is obtained for each reproducing element, and the obtained difference is referred to as a “non-linear component after storage”. FIG. 3 shows an example of a graph relating to the non-linear component after storage.

(46) An absolute value of a difference (difference between before and after storage of the non-linear component) between the “non-linear component at initial stage” and the “non-linear component after storage” for each reproducing element is obtained. FIG. 4 is an example of a graph

showing the absolute value of the difference obtained for each reproducing element. The maximum value of the absolute value in the tape outer peripheral region and the tape inner peripheral region is defined as a “non-linear component of tape width deformation” of the magnetic tape to be measured. It is considered that the non-linear component at the initial stage can be said to be a non-linear component generated by a factor other than the magnetic tape. Therefore, the present inventor considers that it is possible to accurately evaluate the non-linear component caused by the magnetic tape by obtaining the difference between before and after storage of the non-linear component as described above.

(47) From the viewpoint of improving the operational stability of the drive in recording and/or reproduction after long-term storage, the non-linear component of the tape width deformation obtained by the method described above for the magnetic tape is preferably 100 nm or less, more preferably 95 nm or less, and still more preferably 90 nm or less, 85 nm or less, 80 nm or less, 75 nm or less, 70 nm or less, 65 nm or less, 60 nm or less, 55 nm or less, and 50 nm or less in this order. The V_{vc} of the magnetic layer surface and the width direction σ of the V_{vc} being in the above-described range can contribute to controlling the value of the non-linear component in the above-described range. In addition, the non-linear component of the tape width deformation may be, for example, 0 nm or more, more than 0 nm, 1 nm or more, 5 nm or more, 10 nm or more, 15 nm or more, 20 nm or more, 25 nm or more, 30 nm or more, 35 nm or more, or 40 nm or more. From the viewpoint of improving the operational stability of the drive, a smaller non-linear component of the tape width deformation is preferable.

(48) Description of Head Tilt Angle

(49) As described above, one example of the control means for the dynamic track position during running of the magnetic tape is to change the head tilt angle. Regarding this point, a configuration of the magnetic head, the head tilt angle, and the like will be described below. Further, the reason why it is possible to control the dynamic track position during running of the magnetic tape by tilting the axial direction of the module of the magnetic head against the width direction of the magnetic tape during running of the magnetic tape will also be described below.

(50) The magnetic head may have one or more modules including an element array with a plurality of magnetic head elements between a pair of servo signal reading elements, and may have two or more or three or more modules. The total number of such modules may be, for example, 5 or less, 4 or less, or 3 or less, and the modules as many as the number exceeding the total number illustrated here may be included in the magnetic head. Arrangement examples of a plurality of modules include “recording module–reproducing module” (total number of modules: 2), and “recording module–reproducing module–recording module” (total number of modules: 3). Note that the present invention is not limited to the examples shown here.

(51) Each module can include an element array with a plurality of magnetic head elements between a pair of servo signal reading elements, that is, an arrangement of the elements. A module having a recording element as the magnetic head element is a recording module for recording data on the magnetic tape. A module having a reproducing element as the magnetic head element is a reproducing module for reproducing data recorded on the magnetic tape. In the magnetic head, a plurality of modules are arranged, for example, in a recording and reproducing head unit such that axes of the element arrays of the respective modules are oriented in parallel. Such a term “parallel” does not necessarily mean only parallel in a strict sense, but includes a range of errors normally allowed in the technical field to which the present invention belongs. The range of errors can mean, for example, a range less than strictly parallel $\pm 10^\circ$.

(52) In each element array, the pair of servo signal reading elements and the plurality of magnetic head elements (that is, the recording element or the reproducing element) are usually arranged linearly to be spaced from each other. Here, the term “arranged linearly” means that each magnetic head element is arranged on a straight line connecting a central portion of one servo signal reading element and a central portion of the other servo signal reading element. The term “axis of the

element array” in the present invention and the present specification means a straight line connecting a central portion of one servo signal reading element and a central portion of the other servo signal reading element.

(53) Next, a configuration of a module and the like will be further described with reference to the drawings. Note that the form shown in the drawings is an example and does not limit the present invention.

(54) FIG. 5 is a schematic view showing an example of the module of the magnetic head. The module shown in FIG. 5 has a plurality of magnetic head elements between a pair of servo signal reading elements (servo signal reading elements **1** and **2**). The magnetic head element is also referred to as a “channel”. “Ch” in the figure is an abbreviation for channel. The module shown in FIG. 5 has a total of 32 magnetic head elements from Ch**0** to Ch**31**.

(55) In FIG. 5, “L” represents a distance between a pair of servo signal reading elements, that is, a distance between one servo signal reading element and the other servo signal reading element. In the module shown in FIG. 5, “L” represents a distance between the servo signal reading element **1** and the servo signal reading element **2**. Specifically, it is a distance between a central portion of the servo signal reading element **1** and a central portion of the servo signal reading element **2**. Such a distance can be measured, for example, by an optical microscope or the like.

(56) FIG. 6 is an explanatory diagram of a relative positional relationship between the module and the magnetic tape during running of the magnetic tape in the magnetic tape apparatus. In FIG. 6, the dotted line A indicates the width direction of the magnetic tape. The dotted line B indicates the axis of the element array. The angle θ can be said to be a head tilt angle during running of the magnetic tape, and is an angle formed by the dotted line A and the dotted line B. In a case where the angle θ is 0° during running of the magnetic tape, a distance in the magnetic tape width direction between one servo signal reading element and the other servo signal reading element in the element array (hereinafter, also referred to as “effective distance between servo signal reading elements”) is “L”. On the other hand, in a case where the angle θ exceeds 0° , the effective distance between the servo signal reading elements is “ $L\cos\theta$ ”, where $L\cos\theta$ is smaller than L. That is, “ $L\cos\theta < L$ ”.

(57) As described above, during recording or reproduction, in a case where the magnetic head for recording or reproducing data records or reproduces data while being deviated from a target track position due to the width deformation of the magnetic tape, a phenomenon such as overwriting of recorded data or reproduction failure may occur. For example, in a case where the width of the magnetic tape contracts or expands, a phenomenon may occur in which the magnetic head element, which should perform recording or reproduction at a target track position, performs recording or reproduction at a different track position. In addition, in a case where the width of the magnetic tape expands, a phenomenon may occur in which the effective distance between the servo signal reading elements becomes shorter than an interval between two servo bands adjacent to each other with the data band interposed therebetween (also referred to as “servo band interval” or “interval between servo bands”, specifically, a distance between the two servo bands in the width direction of the magnetic tape), and data is not recorded or reproduced in a portion near an edge of the magnetic tape.

(58) On the other hand, in a case where the element array is tilted at an angle θ exceeding 0° , the effective distance between the servo signal reading elements becomes “ $L\cos\theta$ ” as described above. The larger the value of θ , the smaller the value of $L\cos\theta$, and the smaller the value of θ , the larger the value of $L\cos\theta$. Therefore, by changing the value of θ according to a degree of the dimension change (that is, contraction or extension) in the width direction of the magnetic tape, it is possible to make the effective distance between the servo signal reading elements approximate to or match with the interval between the servo bands. As a result, it is possible to prevent a phenomenon such as overwriting of recorded data or reproduction failure due to the fact that the magnetic head for recording or reproducing data records or reproduces data while being deviated from a target track

position due to the width deformation of the magnetic tape during recording or reproduction, or to reduce a frequency of the occurrence of the phenomenon.

(59) FIG. 7 is an explanatory diagram relating to a change in the angle θ during running of the magnetic tape.

(60) $\theta_{\text{sub.initial}}$, which is an angle θ at the start of running, can be set to, for example, 0° or more or more than 0° .

(61) In FIG. 7, the central figure shows a state of the module at the start of running.

(62) In FIG. 7, the right figure shows a state of the module in a case where the angle θ is set to an angle $\theta_{\text{sub.c}}$, which is an angle larger than $\theta_{\text{sub.initial}}$. The effective distance between the servo signal reading elements $L\cos\theta_{\text{sub.c}}$ is a value smaller than $L\cos\theta_{\text{sub.initial}}$ at the start of running of the magnetic tape. In a case where the width of the magnetic tape contracts during running of the magnetic tape, it is preferable to perform such angle adjustment.

(63) On the other hand, in FIG. 7, the left figure shows a state of the module in a case where the angle θ is set to an angle $\theta_{\text{sub.e}}$, which is an angle smaller than $\theta_{\text{sub.initial}}$. The effective distance between the servo signal reading elements $L\cos\theta_{\text{sub.e}}$ is a value larger than $L\cos\theta_{\text{sub.initial}}$ at the start of running of the magnetic tape. In a case where the width of the magnetic tape expands during running of the magnetic tape, it is preferable to perform such angle adjustment.

(64) As described above, changing the head tilt angle during running of the magnetic tape can contribute to prevention of the phenomenon such as overwriting of recorded data or reproduction failure due to the fact that the magnetic head for recording or reproducing data records or reproduces data while being deviated from a target track position due to the width deformation of the magnetic tape during recording or reproduction, or can contribute to reduction of the frequency of the occurrence of the phenomenon.

(65) Note that, for example, with the control means for the dynamic track position as described above, it is usually possible to compensate for off-track caused by the linear component of the tape width deformation, but it is difficult to suppress off-track caused by the non-linear component. On the other hand, in the magnetic tape, it is speculated that the V_{vc} of the magnetic layer surface and the width direction σ of the V_{vc} being in the above-described range contribute to the reduction of the non-linear component of the tape width deformation. It is considered that this makes it possible to improve the operational stability of the drive. Such a magnetic tape is preferable in order to further increase track density.

(66) Hereinafter, the magnetic tape will be described in more detail.

(67) Magnetic Layer

(68) Ferromagnetic Powder

(69) As a ferromagnetic powder included in the magnetic layer, a well-known ferromagnetic powder as a ferromagnetic powder used in magnetic layers of various magnetic recording media can be used alone or in combination of two or more. From the viewpoint of improving recording density, it is preferable to use a ferromagnetic powder having a small average particle size. From this point, the average particle size of the ferromagnetic powder is preferably 50 nm or less, more preferably 45 nm or less, still more preferably 40 nm or less, still more preferably 35 nm or less, still more preferably 30 nm or less, still more preferably 25 nm or less, and still more preferably 20 nm or less. On the other hand, from the viewpoint of magnetization stability, the average particle size of the ferromagnetic powder is preferably 5 nm or more, more preferably 8 nm or more, still more preferably 10 nm or more, still more preferably 15 nm or more, and still more preferably 20 nm or more.

(70) Hexagonal Ferrite Powder

(71) Preferred specific examples of the ferromagnetic powder include a hexagonal ferrite powder. For details of the hexagonal ferrite powder, for example, descriptions disclosed in paragraphs 0012 to 0030 of JP2011-225417A, paragraphs 0134 to 0136 of JP2011-216149A, paragraphs 0013 to 0030 of JP2012-204726A, and paragraphs 0029 to 0084 of JP2015-127985A can be referred to.

(72) In the present invention and the present specification, the term “hexagonal ferrite powder” refers to a ferromagnetic powder in which a hexagonal ferrite crystal structure is detected as a main phase by X-ray diffraction analysis. The main phase refers to a structure to which the highest intensity diffraction peak in an X-ray diffraction spectrum obtained by X-ray diffraction analysis is attributed. For example, in a case where the highest intensity diffraction peak is attributed to a hexagonal ferrite crystal structure in an X-ray diffraction spectrum obtained by X-ray diffraction analysis, it is determined that the hexagonal ferrite crystal structure is detected as the main phase. In a case where only a single structure is detected by X-ray diffraction analysis, this detected structure is taken as the main phase. The hexagonal ferrite crystal structure includes at least an iron atom, a divalent metal atom, and an oxygen atom, as a constituent atom. The divalent metal atom is a metal atom that can be a divalent cation as an ion, and examples thereof may include an alkaline earth metal atom such as a strontium atom, a barium atom, and a calcium atom, and a lead atom. In the present invention and the present specification, a hexagonal strontium ferrite powder refers to a powder in which a main divalent metal atom is a strontium atom, and a hexagonal barium ferrite powder refers to a powder in which a main divalent metal atom is a barium atom. The main divalent metal atom refers to a divalent metal atom that accounts for the most on an at % basis among the divalent metal atoms included in the powder. Note that a rare earth atom is not included in the above divalent metal atom. The term “rare earth atom” in the present invention and the present specification is selected from the group consisting of a scandium atom (Sc), an yttrium atom (Y), and a lanthanoid atom. The lanthanoid atom is selected from the group consisting of a lanthanum atom (La), a cerium atom (Ce), a praseodymium atom (Pr), a neodymium atom (Nd), a promethium atom (Pm), a samarium atom (Sm), a europium atom (Eu), a gadolinium atom (Gd), a terbium atom (Tb), a dysprosium atom (Dy), a holmium atom (Ho), an erbium atom (Er), a thulium atom (Tm), an ytterbium atom (Yb), and a lutetium atom (Lu).

(73) Hereinafter, the hexagonal strontium ferrite powder, which is one aspect of the hexagonal ferrite powder, will be described in more detail.

(74) An activation volume of the hexagonal strontium ferrite powder is preferably in a range of 800 to 1600 nm.³. The finely granulated hexagonal strontium ferrite powder having an activation volume in the above range is suitable for manufacturing a magnetic tape exhibiting excellent electromagnetic conversion characteristics. The activation volume of the hexagonal strontium ferrite powder is preferably 800 nm.³ or more, and may be, for example, 850 nm.³ or more. Further, from the viewpoint of further improving the electromagnetic conversion characteristics, the activation volume of the hexagonal strontium ferrite powder is more preferably 1500 nm.³ or less, still more preferably 1400 nm.³ or less, still more preferably 1300 nm.³ or less, still more preferably 1200 nm.³ or less, and still more preferably 1100 nm.³ or less. The same applies to an activation volume of the hexagonal barium ferrite powder.

(75) The term “activation volume” refers to a unit of magnetization reversal and is an index indicating the magnetic size of a particle. An activation volume described in the present invention and the present specification and an anisotropy constant K_u which will be described below are values obtained from the following relational expression between a coercivity H_c and an activation volume V , by performing measurement in a coercivity H_c measurement portion at a magnetic field sweep rate of 3 minutes and 30 minutes using a vibrating sample magnetometer (measurement temperature: 23° C.±1° C.). For a unit of the anisotropy constant K_u , 1 erg/cc=1.0×10.⁻¹ J/m.³.

$$(76) H_c = 2K_u / M_s \{ 1 - [(kT / K_u V) \ln(A\tau / 0.693)]^{1/2} \}$$

(77) [In the above expression, K_u : anisotropy constant (unit: J/m.³), M_s : saturation magnetization (Unit: kA/m), k : Boltzmann constant, T : absolute temperature (unit: K), V : activation volume (unit: cm.³), A : spin precession frequency (unit: s.⁻¹), τ : magnetic field reversal time (unit: s)]

(78) An index for reducing thermal fluctuation, in other words, for improving the thermal stability

may include the anisotropy constant K_u . The hexagonal strontium ferrite powder preferably has K_u of $1.8 \times 10^{5.5}$ J/m³ or more, and more preferably has K_u of $2.0 \times 10^{5.5}$ J/m³ or more. K_u of the hexagonal strontium ferrite powder may be, for example, $2.5 \times 10^{5.5}$ J/m³ or less. Here, since higher K_u means higher thermal stability, which is preferable, a value thereof is not limited to the values exemplified above.

(79) The hexagonal strontium ferrite powder may or may not include a rare earth atom. In a case where the hexagonal strontium ferrite powder includes a rare earth atom, it is preferable to include a rare earth atom at a content (bulk content) of 0.5 to 5.0 at % with respect to 100 at % of an iron atom. In one aspect, the hexagonal strontium ferrite powder including a rare earth atom may have a rare earth atom surface layer portion uneven distribution property. The “rare earth atom surface layer portion uneven distribution property” of the present invention and the present specification means that a content of rare earth atom with respect to 100 at % of iron atom in a solution obtained by partially dissolving the hexagonal strontium ferrite powder with acid (hereinafter, referred to as a “rare earth atom surface layer portion content” or simply a “surface layer portion content” regarding the rare earth atom) and a content of rare earth atom with respect to 100 at % of iron atom in a solution obtained by totally dissolving the hexagonal strontium ferrite powder with acid (hereinafter, referred to as a “rare earth atom bulk content” or simply a “bulk content” regarding the rare earth atom) satisfy a ratio of a rare earth atom surface layer portion content/a rare earth atom bulk content >1.0 . A rare earth atom content in the hexagonal strontium ferrite powder described below is synonymous with the rare earth atom bulk content. On the other hand, partial dissolution using an acid dissolves a surface layer portion of a particle constituting the hexagonal strontium ferrite powder, and thus, a rare earth atom content in a solution obtained by partial dissolution is a rare earth atom content in a surface layer portion of a particle constituting the hexagonal strontium ferrite powder. A rare earth atom surface layer portion content satisfying a ratio of “rare earth atom surface layer portion content/rare earth atom bulk content >1.0 ” means that in a particle constituting the hexagonal strontium ferrite powder, rare earth atoms are unevenly distributed in a surface layer portion (that is, more than an inside). The surface layer portion in the present invention and the present specification means a partial region from a surface of a particle constituting the hexagonal strontium ferrite powder toward an inside.

(80) In a case where the hexagonal strontium ferrite powder includes the rare earth atom, a rare earth atom content (bulk content) is preferably in a range of 0.5 to 5.0 at % with respect to 100 at % of an iron atom. It is considered that a bulk content in the above range of the included rare earth atom and uneven distribution of the rare earth atoms in a surface layer portion of a particle constituting the hexagonal strontium ferrite powder contribute to suppression of a decrease in reproduction output during repeated reproduction. It is supposed that this is because the hexagonal strontium ferrite powder includes a rare earth atom with a bulk content in the above range, and rare earth atoms are unevenly distributed in a surface layer portion of a particle constituting the hexagonal strontium ferrite powder, and thus it is possible to increase an anisotropy constant K_u . The higher a value of an anisotropy constant K_u is, the more a phenomenon called thermal fluctuation can be suppressed (in other words, thermal stability can be improved). By suppressing occurrence of thermal fluctuation, it is possible to suppress a decrease in reproduction output during repeated reproduction. It is supposed that uneven distribution of rare earth atoms in a particulate surface layer portion of the hexagonal strontium ferrite powder contributes to stabilization of spins of iron (Fe) sites in a crystal lattice of a surface layer portion, and thus, an anisotropy constant K_u may be increased.

(81) Moreover, it is supposed that the use of the hexagonal strontium ferrite powder having a rare earth atom surface layer portion uneven distribution property as a ferromagnetic powder in the magnetic layer also contributes to inhibition of a magnetic layer surface from being scraped by being slid with respect to the magnetic head. That is, it is supposed that the hexagonal strontium ferrite powder having a rare earth atom surface layer portion uneven distribution property can also

contribute to an improvement of running durability of the magnetic tape. It is supposed that this may be because uneven distribution of rare earth atoms on a surface of a particle constituting the hexagonal strontium ferrite powder contributes to an improvement of interaction between the particle surface and an organic substance (for example, a binding agent and/or an additive) included in the magnetic layer, and, as a result, a strength of the magnetic layer is improved.

(82) From the viewpoint of further suppressing a decrease in reproduction output during repeated reproduction and/or the viewpoint of further improving running durability, the rare earth atom content (bulk content) is more preferably in a range of 0.5 to 4.5 at %, still more preferably in a range of 1.0 to 4.5 at %, and still more preferably in a range of 1.5 to 4.5 at %.

(83) The bulk content is a content obtained by totally dissolving the hexagonal strontium ferrite powder. In the present invention and the present specification, unless otherwise noted, the content of an atom means a bulk content obtained by totally dissolving the hexagonal strontium ferrite powder. The hexagonal strontium ferrite powder including a rare earth atom may include only one kind of rare earth atom as the rare earth atom, or may include two or more kinds of rare earth atoms. The bulk content in a case of including two or more kinds of rare earth atoms is obtained for the total of two or more kinds of rare earth atoms. This also applies to other components in the present invention and the present specification. That is, unless otherwise noted, a certain component may be used alone or in combination of two or more. A content amount or a content in a case where two or more components are used refers to that for the total of two or more components.

(84) In a case where the hexagonal strontium ferrite powder includes a rare earth atom, the included rare earth atom need only be any one or more of rare earth atoms. As a rare earth atom that is preferable from the viewpoint of further suppressing a decrease in reproduction output during repeated reproduction, there are a neodymium atom, a samarium atom, a yttrium atom, and a dysprosium atom, here, the neodymium atom, the samarium atom, and the yttrium atom are more preferable, and a neodymium atom is still more preferable.

(85) In the hexagonal strontium ferrite powder having a rare earth atom surface layer portion uneven distribution property, the rare earth atoms need only be unevenly distributed in the surface layer portion of a particle constituting the hexagonal strontium ferrite powder, and the degree of uneven distribution is not limited. For example, for the hexagonal strontium ferrite powder having a rare earth atom surface layer portion uneven distribution property, a ratio of a surface layer portion content of a rare earth atom obtained by partial dissolution under dissolution conditions which will be described below to a bulk content of a rare earth atom obtained by total dissolution under dissolution conditions which will be described below, that is, “surface layer portion content/bulk content” exceeds 1.0 and may be 1.5 or more. The fact that “surface layer portion content/bulk content” is larger than 1.0 means that in a particle constituting the hexagonal strontium ferrite powder, rare earth atoms are unevenly distributed in the surface layer portion (that is, more than an inside). Further, a ratio of a surface layer portion content of a rare earth atom obtained by partial dissolution under dissolution conditions which will be described below to a bulk content of a rare earth atom obtained by total dissolution under the dissolution conditions which will be described below, that is, “surface layer portion content/bulk content” may be, for example, 10.0 or less, 9.0 or less, 8.0 or less, 7.0 or less, 6.0 or less, 5.0 or less, or 4.0 or less. Note that, in the hexagonal strontium ferrite powder having a rare earth atom surface layer portion uneven distribution property, the rare earth atoms need only be unevenly distributed in the surface layer portion of a particle constituting the hexagonal strontium ferrite powder, and the “surface layer portion content/bulk content” is not limited to the exemplified upper limit or lower limit.

(86) The partial dissolution and the total dissolution of the hexagonal strontium ferrite powder will be described below. For the hexagonal strontium ferrite powder that exists as a powder, the partially and totally dissolved sample powder is taken from the same lot of powder. On the other hand, for the hexagonal strontium ferrite powder included in the magnetic layer of the magnetic

tape, a part of the hexagonal strontium ferrite powder taken out from the magnetic layer is subjected to partial dissolution, and the other part is subjected to total dissolution. The hexagonal strontium ferrite powder can be taken out from the magnetic layer by a method described in a paragraph 0032 of JP2015-91747A, for example.

(87) The partial dissolution means that dissolution is performed such that, at the end of dissolution, the residue of the hexagonal strontium ferrite powder can be visually checked in the solution. For example, by partial dissolution, it is possible to dissolve a region of 10 to 20 mass % of the particle constituting the hexagonal strontium ferrite powder with the total particle being 100 mass %. On the other hand, the total dissolution means that dissolution is performed such that, at the end of dissolution, the residue of the hexagonal strontium ferrite powder cannot be visually checked in the solution.

(88) The partial dissolution and measurement of the surface layer portion content are performed by the following method, for example. Note that the following dissolution conditions such as the amount of sample powder are exemplified, and dissolution conditions for partial dissolution and total dissolution can be employed in any manner.

(89) A container (for example, a beaker) containing 12 mg of the sample powder and 10 mL of 1 mol/L hydrochloric acid is held on a hot plate at a set temperature of 70° C. for 1 hour. The obtained solution is filtered by a membrane filter of 0.1 μm . Elemental analysis of the filtrated solution thus obtained is performed by an inductively coupled plasma (ICP) analyzer. In this way, the surface layer portion content of a rare earth atom with respect to 100 at % of an iron atom can be obtained. In a case where a plurality of kinds of rare earth atoms are detected by elemental analysis, the total content of all rare earth atoms is defined as the surface layer portion content. This also applies to the measurement of the bulk content.

(90) On the other hand, the total dissolution and measurement of the bulk content are performed by the following method, for example.

(91) A container (for example, a beaker) containing 12 mg of the sample powder and 10 mL of 4 mol/L hydrochloric acid is held on a hot plate at a set temperature of 80° C. for 3 hours. Thereafter, the same procedure as the partial dissolution and the measurement of the surface layer portion content is carried out, and the bulk content with respect to 100 at % of an iron atom can be obtained.

(92) From the viewpoint of increasing the reproduction output in a case of reproducing data recorded on the magnetic tape, it is desirable that mass magnetization as of the ferromagnetic powder included in the magnetic tape is high. In this regard, the hexagonal strontium ferrite powder including a rare earth atom but not having the rare earth atom surface layer portion uneven distribution property tends to have a larger decrease in σ_s than that of the hexagonal strontium ferrite powder including no rare earth atom. With respect to this, it is considered that the hexagonal strontium ferrite powder having a rare earth atom surface layer portion uneven distribution property is preferable in suppressing such a large decrease in σ_s . In one aspect, as of the hexagonal strontium ferrite powder may be 45 A.Math.m.sup.2/kg or more, and may be 47 A.Math.m.sup.2/kg or more. On the other hand, from the viewpoint of noise reduction, σ_s is preferably 80 A.Math.m.sup.2/kg or less and more preferably 60 A.Math.m.sup.2/kg or less. σ_s can be measured using a well-known measuring device, such as a vibrating sample magnetometer, capable of measuring magnetic properties. In the present invention and the present specification, unless otherwise noted, the mass magnetization as is a value measured at a magnetic field intensity of 15 kOe. 1 [kOe] is $10 \cdot 10^3 / 4\pi$ [A/m].

(93) Regarding the content (bulk content) of a constituent atom of the hexagonal strontium ferrite powder, a strontium atom content may be, for example, in a range of 2.0 to 15.0 at % with respect to 100 at % of an iron atom. In one aspect, the hexagonal strontium ferrite powder may include only a strontium atom as a divalent metal atom. In another aspect, the hexagonal strontium ferrite powder may include one or more other divalent metal atoms in addition to a strontium atom. For

example, a barium atom and/or a calcium atom may be included. In a case where the other divalent metal atoms other than the strontium atom are included, a content of the barium atom and a content of the calcium atom in the hexagonal strontium ferrite powder respectively can be, for example, in a range of 0.05 to 5.0 at % with respect to 100 at % of the iron atom.

(94) As the hexagonal ferrite crystal structure, a magnetoplumbite type (also referred to as an “M type”), a W type, a Y type, and a Z type are known. The hexagonal strontium ferrite powder may have any crystal structure. The crystal structure can be checked by X-ray diffraction analysis. In the hexagonal strontium ferrite powder, a single crystal structure or two or more crystal structures may be detected by X-ray diffraction analysis. For example, according to one aspect, in the hexagonal strontium ferrite powder, only the M-type crystal structure may be detected by X-ray diffraction analysis. For example, M type hexagonal ferrite is represented by a composition formula of $A\text{Fe}_{12}\text{O}_{19}$. Here, A represents a divalent metal atom, and in a case where the hexagonal strontium ferrite powder is the M type, A is only a strontium atom (Sr), or in a case where, as A, a plurality of divalent metal atoms are included, as described above, a strontium atom (Sr) accounts for the most on an at % basis. The divalent metal atom content of the hexagonal strontium ferrite powder is usually determined by the type of crystal structure of the hexagonal ferrite and is not particularly limited. The same applies to the iron atom content and the oxygen atom content. The hexagonal strontium ferrite powder may include at least an iron atom, a strontium atom, and an oxygen atom, and may further include a rare earth atom. Furthermore, the hexagonal strontium ferrite powder may or may not include atoms other than these atoms. As an example, the hexagonal strontium ferrite powder may include an aluminum atom (Al). A content of an aluminum atom may be, for example, 0.5 to 10.0 at % with respect to 100 at % of an iron atom. From the viewpoint of further suppressing a decrease in reproduction output during repeated reproduction, the hexagonal strontium ferrite powder includes an iron atom, a strontium atom, an oxygen atom, and a rare earth atom, and the content of atoms other than these atoms is preferably 10.0 at % or less, more preferably in a range of 0 to 5.0 at %, and may be 0 at % with respect to 100 at % of an iron atom. That is, in one aspect, the hexagonal strontium ferrite powder may not include atoms other than an iron atom, a strontium atom, an oxygen atom, and a rare earth atom. The content expressed in at % is obtained by converting a content of each atom (unit: mass %) obtained by totally dissolving the hexagonal strontium ferrite powder into a value expressed in at % using an atomic weight of each atom. Further, in the present invention and the present specification, the term “not include” for a certain atom means that a content measured by an ICP analyzer after total dissolution is 0 mass %. A detection limit of the ICP analyzer is usually 0.01 parts per million (ppm) or less on a mass basis. The term “not included” is used as a meaning including that an atom is included in an amount less than the detection limit of the ICP analyzer. In one aspect, the hexagonal strontium ferrite powder may not include a bismuth atom (Bi).

(95) Metal Powder

(96) Preferred specific examples of the ferromagnetic powder include a ferromagnetic metal powder. For details of the ferromagnetic metal powder, descriptions disclosed in paragraphs 0137 to 0141 of JP2011-216149A and paragraphs 0009 to 0023 of JP2005-251351A can be referred to, for example.

(97) ϵ -Iron Oxide Powder

(98) Preferred specific examples of the ferromagnetic powder include an ϵ -iron oxide powder. In the present invention and the present specification, the term “ ϵ -iron oxide powder” refers to a ferromagnetic powder in which an ϵ -iron oxide crystal structure is detected as a main phase by X-ray diffraction analysis. For example, in a case where the highest intensity diffraction peak is attributed to an ϵ -iron oxide crystal structure in an X-ray diffraction spectrum obtained by X-ray diffraction analysis, it is determined that the ϵ -iron oxide crystal structure is detected as the main phase. As a method of manufacturing an ϵ -iron oxide powder, a manufacturing method from a goethite, a reverse micelle method, and the like are known. All of the manufacturing methods are

well known. Regarding a method of manufacturing an ϵ -iron oxide powder in which a part of Fe is substituted with substitutional atoms such as Ga, Co, Ti, Al, or Rh, a description disclosed in J. Jpn. Soc. Powder Metallurgy Vol. 61 Supplement, No. S1, pp. S280 to S284, J. Mater. Chem. C, 2013, 1, pp. 5200 to 5206 can be referred to, for example. Note that the method of manufacturing the ϵ -iron oxide powder capable of being used as the ferromagnetic powder in the magnetic layer of the magnetic tape is not limited to the methods described here.

(99) An activation volume of the ϵ -iron oxide powder is preferably in a range of 300 to 1500 nm.³. The finely granulated ϵ -iron oxide powder having an activation volume in the above range is suitable for manufacturing a magnetic tape exhibiting excellent electromagnetic conversion characteristics. The activation volume of the ϵ -iron oxide powder is preferably 300 nm.³ or more, and may be, for example, 500 nm.³ or more. Further, from the viewpoint of further improving the electromagnetic conversion characteristics, the activation volume of the ϵ -iron oxide powder is more preferably 1400 nm.³ or less, still more preferably 1300 nm.³ or less, still more preferably 1200 nm.³ or less, and still more preferably 1100 nm.³ or less.

(100) An index for reducing thermal fluctuation, in other words, for improving the thermal stability may include the anisotropy constant K_u . The ϵ -iron oxide powder preferably has K_u of 3.0×10^4 J/m.³ or more, and more preferably has K_u of 8.0×10^4 J/m.³ or more. K_u of the ϵ -iron oxide powder may be, for example, 3.0×10^5 J/m.³ or less. Here, since higher K_u means higher thermal stability, which is preferable, a value thereof is not limited to the values exemplified above.

(101) From the viewpoint of increasing the reproduction output in a case of reproducing data recorded on the magnetic tape, it is desirable that mass magnetization as of the ferromagnetic powder included in the magnetic tape is high. In this regard, in one aspect, as of the ϵ -iron oxide powder may be 8 A.Math.m.²/kg or more, and may be 12 A.Math.m.²/kg or more. On the other hand, from the viewpoint of noise reduction, as of the ϵ -iron oxide powder is preferably 40 A.Math.m.²/kg or less and more preferably 35 A.Math.m.²/kg or less.

(102) In the present invention and the present specification, unless otherwise noted, an average particle size of various powders such as ferromagnetic powders is a value measured by the following method using a transmission electron microscope.

(103) The powder is imaged at an imaging magnification of 100000 $\times$ with a transmission electron microscope, the image is printed on photographic printing paper or displayed on a display so that the total magnification of 500000 $\times$ to obtain an image of particles configuring the powder. A target particle is selected from the obtained image of particles, an outline of the particle is traced by a digitizer, and a size of the particle (primary particle) is measured. The primary particles are independent particles without aggregation.

(104) The measurement described above is performed regarding 500 particles randomly extracted. An arithmetic average of the particle sizes of 500 particles thus obtained is an average particle size of the powder. As the transmission electron microscope, a transmission electron microscope H-9000 manufactured by Hitachi, Ltd. can be used, for example. In addition, the measurement of the particle size can be performed by well-known image analysis software, for example, image analysis software KS-400 manufactured by Carl Zeiss. An average particle size described in the section of Examples which will be described below is a value measured by using a transmission electron microscope H-9000 manufactured by Hitachi, Ltd. as the transmission electron microscope, and image analysis software KS-400 manufactured by Carl Zeiss as the image analysis software, unless otherwise noted. In the present invention and the present specification, the powder means an aggregate of a plurality of particles. For example, the ferromagnetic powder means an aggregate of a plurality of ferromagnetic particles. Further, the aggregate of the plurality of particles not only includes an aspect in which particles constituting the aggregate directly come into contact with each other, but also includes an aspect in which a binding agent or an additive which will be described below is interposed between the particles. The term "particle" is used to describe a

powder in some cases.

(105) As a method of taking a sample powder from the magnetic tape in order to measure the particle size, a method disclosed in a paragraph 0015 of JP2011-048878A can be employed, for example.

(106) In the present invention and the present specification, regarding the shape of the particle, the “plate shape” means a shape in which two plate surfaces face each other. The term “plate surface” refers to a plane observed on the surface of the particle of the particle image described above and a plane facing the plane. On the other hand, among the particle shapes that do not have such plate surfaces, the shape in which a long axis and a short axis are distinguished is the “elliptical shape”. The long axis is determined as an axis (the straight line) along which the longest particle length can be taken. On the other hand, the short axis is determined as an axis along which the particle length is longest in a case where the particle length is taken along a straight line orthogonal to the long axis. A shape in which the long axis and the short axis are not distinguished, that is, a shape in which the long axis length is equal to the short axis length is the “spherical shape”. A shape in which the long axis and the short axis cannot be specified from the shape is called an amorphous shape.

(107) In the present invention and the present specification, unless otherwise noted, the size (particle size) of the particles constituting the powder is a plate diameter in a case where the shape of the particles observed in the particle image is plate-shaped, is a long axis length in a case where the shape is elliptical, is a diameter in a case where the shape is spherical, and is an equivalent circle diameter in a case where the shape is amorphous. The equivalent circle diameter is obtained by a circle projection method.

(108) Regarding the plate-shaped particles, the “plate thickness” is the longest distance between two facing plate surfaces, and the average plate thickness is the arithmetic average of values obtained for the 500 particles.

(109) The content (filling percentage) of the ferromagnetic powder of the magnetic layer is preferably in a range of 50 to 90 mass % and more preferably in a range of 60 to 90 mass %, with respect to the total mass of the magnetic layer. A high filling percentage of the ferromagnetic powder in the magnetic layer is preferable from the viewpoint of improving the recording density.

(110) Binding Agent

(111) The magnetic tape can be a coating type magnetic tape, and include a binding agent in the magnetic layer. The binding agent is one or more resins. As the binding agent, various resins usually used as a binding agent of a coating type magnetic tape can be used. For example, as the binding agent, a resin selected from a polyurethane resin, a polyester resin, a polyamide resin, a vinyl chloride resin, an acrylic resin obtained by copolymerizing styrene, acrylonitrile, or methyl methacrylate, a cellulose resin such as nitrocellulose, an epoxy resin, a phenoxy resin, and a polyvinylalkylal resin such as polyvinyl acetal or polyvinyl butyral can be used alone or a plurality of resins can be mixed with each other to be used. Among these, a polyurethane resin, an acrylic resin, a cellulose resin, and a vinyl chloride resin are preferable. These resins may be homopolymers or copolymers. These resins can be used as the binding agent even in a non-magnetic layer and/or a back coating layer which will be described below.

(112) For the above binding agent, descriptions disclosed in paragraphs 0028 to 0031 of JP2010-24113A can be referred to. An average molecular weight of the resin used as the binding agent can be, for example, 10,000 to 200,000 as a weight-average molecular weight. The binding agent may be used in an amount of, for example, 1.0 to 30.0 parts by mass with respect to 100.0 parts by mass of the ferromagnetic powder.

(113) Curing Agent

(114) A curing agent can also be used together with the resin which can be used as the binding agent. As the curing agent, in one aspect, a thermosetting compound which is a compound in which curing reaction (crosslinking reaction) proceeds due to heating can be used, and in another aspect, a

photocurable compound in which a curing reaction (crosslinking reaction) proceeds due to light irradiation can be used. The curing reaction proceeds in a magnetic layer forming step, whereby at least a part of the curing agent can be included in the magnetic layer in a state of being reacted (crosslinked) with other components such as the binding agent. The same applies to the layer formed using this composition in a case where the composition used to form the other layer includes a curing agent. The preferred curing agent is a thermosetting compound, and polyisocyanate is suitable for this. For details of the polyisocyanate, descriptions disclosed in paragraphs 0124 and 0125 of JP2011-216149A can be referred to. The curing agent can be used in the composition for forming a magnetic layer in an amount of, for example, 0 to 80.0 parts by mass, and preferably 50.0 to 80.0 parts by mass from the viewpoint of improving a strength of the magnetic layer, with respect to 100.0 parts by mass of the binding agent.

(115) Additive

(116) The magnetic layer may include one or more kinds of additives, as necessary. As the additive, a commercially available product can be appropriately selected and used according to a desired property. Alternatively, a compound synthesized by a well-known method can be used as the additive. The additive can be used in any amount. Examples of the additive include the curing agent described above. In addition, examples of the additive which can be included in the magnetic layer include a non-magnetic powder (for example, an inorganic powder or carbon black), a lubricant, a dispersing agent, a dispersing assistant, an antibacterial agent, an antistatic agent, an antioxidant, and the like. For example, for the lubricant, descriptions disclosed in paragraphs 0030 to 0033, 0035, and 0036 of JP2016-126817A can be referred to. The non-magnetic layer which will be described below may contain a lubricant. For the lubricant that can be contained in the non-magnetic layer, descriptions disclosed in paragraphs 0030, 0031, 0034, 0035, and 0036 of JP2016-126817A can be referred to. For the dispersing agent, descriptions disclosed in paragraphs 0061 and 0071 of JP2012-133837A can be referred to. The dispersing agent may be added to a composition for forming a non-magnetic layer. For the dispersing agent that can be added to the composition for forming a non-magnetic layer, a description disclosed in a paragraph 0061 of JP2012-133837A can be referred to. As the non-magnetic powder that can be contained in the magnetic layer, a non-magnetic powder which can function as an abrasive, or a non-magnetic powder which can function as a protrusion forming agent which forms protrusions appropriately protruded from the magnetic layer surface (for example, non-magnetic colloidal particles) is used. For example, for the abrasive, descriptions disclosed in paragraphs 0030 to 0032 of JP2004-273070A can be referred to. As protrusion forming agent, the colloidal particles are preferable, and, from the viewpoint of availability, inorganic colloidal particles are preferable, inorganic oxide colloidal particles are more preferable, and silica colloidal particles (colloidal silica) are still more preferable. An average particle size of each of the abrasive and the protrusion forming agent is preferably in a range of 30 to 200 nm, and more preferably in a range of 50 to 100 nm.

(117) The magnetic layer described above can be provided on a surface of the non-magnetic support directly or indirectly through the non-magnetic layer.

(118) Non-Magnetic Layer

(119) Next, the non-magnetic layer will be described. The above magnetic tape may have a magnetic layer directly on the non-magnetic support, or may have a non-magnetic layer including a non-magnetic powder between the non-magnetic support and the magnetic layer. The non-magnetic powder used for the non-magnetic layer may be an inorganic substance powder (inorganic powder) or an organic substance powder (organic powder). In addition, the carbon black and the like can be used. Examples of the inorganic substance include metal, metal oxide, metal carbonate, metal sulfate, metal nitride, metal carbide, and metal sulfide. The non-magnetic powder can be purchased as a commercially available product or can be manufactured by a well-known method. For details thereof, descriptions disclosed in paragraphs 0146 to 0150 of JP2011-216149A can be referred to. For carbon black which can be used in the non-magnetic layer, descriptions disclosed in paragraphs

0040 and 0041 of JP2010-24113A can be referred to. The content (filling percentage) of the non-magnetic powder of the non-magnetic layer is preferably in a range of 50 to 90 mass % and more preferably in a range of 60 to 90 mass %, with respect to the total mass of the non-magnetic layer. (120) In one aspect, the non-magnetic layer can contain an Fe-based inorganic oxide powder as the non-magnetic powder. In the present invention and the present specification, the term “Fe-based inorganic oxide powder” refers to an inorganic oxide powder containing iron as a constituent element. Specific examples of the Fe-based inorganic oxide powder include an α -iron oxide powder and a goethite powder. In the present invention and the present specification, the term “ α -iron oxide powder” refers to a non-magnetic powder in which an α -iron oxide crystal structure is detected as a main phase by X-ray diffraction analysis. The α -iron oxide powder is also generally called hematite or the like.

(121) From the viewpoint of controlling the width direction σ of the V_{vc} within the range described above, it was clarified from the studies of the present inventor that it is preferable that an Fe-based inorganic oxide powder having an average particle volume of $2.0 \times 10^{-6} \mu\text{m}^3$ or less is used as the non-magnetic powder of the non-magnetic layer. Therefore, the average particle volume of the Fe-based inorganic oxide powder included in the non-magnetic layer is preferably $2.0 \times 10^{-6} \mu\text{m}^3$ or less, more preferably $1.5 \times 10^{-6} \mu\text{m}^3$ or less, and still more preferably $1.0 \times 10^{-6} \mu\text{m}^3$ or less. The average particle volume may be, for example, $1.0 \times 10^{-9} \mu\text{m}^3$ or more or $1.0 \times 10^{-8} \mu\text{m}^3$ or more, or may be smaller than the values exemplified here.

(122) In the present invention and the present specification, the average particle volume is a value obtained by the following method.

(123) In order to observe the Fe-based inorganic oxide powder included in the non-magnetic layer of the magnetic tape, first, as a sample pretreatment, slicing is performed by a microtome method. The slicing is performed such that a slicing sample capable of observing a cross section in a thickness direction of the magnetic tape along the longitudinal direction of the magnetic tape can be obtained. For each of Examples and Comparative Examples described later, a slicing sample was obtained using Leica EM UC6 manufactured by Leica Biosystems Nussloch GmbH as a microtome from one of the manufactured magnetic tapes.

(124) A cross section of the obtained slicing sample is observed using a transmission electron microscope (TEM) at an acceleration voltage of 300 kV and a total magnification of $200,000\times$ such that a range from the non-magnetic support to the magnetic layer is included. Thereby, a cross-sectional TEM image is obtained. As the transmission electron microscope, for example, JEM-2100Plus manufactured by JEOL Ltd. can be used. In Examples and Comparative Examples described below, JEM-2100Plus manufactured by JEOL Ltd. was used as a transmission electron microscope.

(125) In the obtained cross-sectional TEM image, 50 particles of the Fe-based inorganic oxide powder are specified with respect to the particles contained in the non-magnetic layer by using a microelectron beam diffraction method. The electron beam diffraction in the microelectron beam diffraction method is performed using a transmission electron microscope at an acceleration voltage of 200 kV and a camera length of 50 cm. In Examples and Comparative Examples described below, in order to perform the electron beam diffraction in the microelectron beam diffraction method, JEM-2100Plus manufactured by JEOL Ltd. was used as a transmission electron microscope.

(126) After that, using 50 particles of the Fe-based inorganic oxide powder specified as described above, the average particle volume is obtained as follows.

(127) First, a long axis length (hereinafter, also referred to as “DL”) and a short axis length (hereinafter, also referred to as “DS”) of each particle are measured.

(128) The long axis length DL means the maximum of distances between two parallel lines drawn from all angles so as to be in contact with the contour of the particle (so-called maximum Feret's

diameter).

(129) In a case where a direction of the long axis length defined as described above is called a long axis direction, the short axis length DS means the maximum of lengths of the particles in a direction orthogonal to the long axis direction of the particles.

(130) Next, an average long axis length DLave is obtained as an arithmetic average of the long axis lengths DL of the 50 measured particles. ave is an abbreviation for average.

(131) In addition, an average short axis length DSave is obtained as an arithmetic average of the short axis lengths DS of the 50 measured particles.

(132) From the average long axis length DLave and the average short axis length DSave, an average particle volume Vave is obtained by the following formula.

(133) $Vave = \pi / 6 \times DSave^2 \times DLave$

(134) In addition, in one aspect, the non-magnetic layer can contain carbon black as the non-magnetic powder. An average particle size of the carbon black can be, for example, 10 nm or more and 50 nm or less. From the viewpoint of controlling the width direction σ of the Vvc within the range described above, it was clarified from the studies of the present inventor that it is preferable that carbon black having a pH of 9.0 or less is used as the non-magnetic powder of the non-magnetic layer. Therefore, the pH of the carbon black contained in the non-magnetic layer is preferably 9.0 or less, more preferably 8.5 or less, and still more preferably 8.0 or less. The pH may be, for example, 1.0 or more, 2.0 or more, 3.0 or more, 4.0 or more, 5.0 or more, or 6.0 or more, or may be smaller than the values exemplified here.

(135) In the present invention and the present specification, the pH of the carbon black is a value measured according to a standard test method ASTM D1512.

(136) The non-magnetic layer preferably contains at least one of an Fe-based inorganic oxide powder having an average particle volume of $2.0 \times 10^{-6} \mu\text{m}^3$ or less or carbon black having a pH of 9.0 or less, and more preferably contains both. A content of an Fe-based inorganic oxide powder having an average particle volume of $2.0 \times 10^{-6} \mu\text{m}^3$ or less with respect to 100.0 parts by mass of the total amount of the non-magnetic powder contained in the non-magnetic layer may be 50.0 parts by mass or more, 60.0 parts by mass or more, or 70.0 parts by mass or more, and for example, 90.0 parts by mass or less. A content of carbon black having a pH of 9.0 or less with respect to 100.0 parts by mass of the total amount of the non-magnetic powder contained in the non-magnetic layer may be 10.0 parts by mass or more or 20.0 parts by mass or more, and for example, 50.0 parts by mass or less, 40.0 parts by mass or less, or 30.0 parts by mass or less.

(137) The non-magnetic layer can include a binding agent, and can also include an additive. For other details of the binding agent or the additive of the non-magnetic layer, a well-known technology regarding the non-magnetic layer can be applied. In addition, in regards to the type and the content of the binding agent, and the type and the content of the additive, for example, the well-known technology regarding the magnetic layer can be applied.

(138) The non-magnetic layer of the magnetic tape also includes a substantially non-magnetic layer including a small amount of ferromagnetic powder as impurities, for example, or intentionally, together with the non-magnetic powder. Here, the substantially non-magnetic layer is a layer having a residual magnetic flux density equal to or smaller than 10 mT, a layer having a coercivity equal to or smaller than 7.96 kA/m (100 Oe), or a layer having a residual magnetic flux density equal to or smaller than 10 mT and a coercivity equal to or smaller than 7.96 kA/m (100 Oe). It is preferable that the non-magnetic layer does not have a residual magnetic flux density and coercivity.

(139) Non-Magnetic Support

(140) Next, the non-magnetic support will be described. Examples of the non-magnetic support (hereinafter, also simply referred to as a “support”) include well-known components such as polyethylene terephthalate, polyethylene naphthalate, polyamide, polyamideimide, and aromatic polyamide subjected to biaxial stretching. Among these, polyethylene terephthalate, polyethylene

naphthalate, and polyamide are preferable. A corona discharge, a plasma treatment, an easy-bonding treatment, or a heat treatment may be performed on these supports in advance.

(141) Back Coating Layer

(142) The magnetic tape may or may not have a back coating layer including a non-magnetic powder on a surface side of the non-magnetic support opposite to a surface side having the magnetic layer. For the non-magnetic powder of the back coating layer, for example, the above description regarding the non-magnetic powder in the non-magnetic layer can be referred to.

(143) The surface shape of the magnetic layer can be affected by transferring a surface shape of a back surface to the magnetic layer surface (so-called show-through) in a state in which the magnetic layer surface and the back surface are in contact in a rolled state in the manufacturing step of the magnetic tape. The back surface is a back coating layer surface in a case where the back coating layer is provided, and is a support surface in a case where the back coating layer is not provided. Regarding the Vvc of the magnetic layer surface and the width direction σ of the Vvc, as a result of the studies of the present inventor, it was clarified that it is preferable that the non-magnetic powder of the back coating layer is a non-magnetic powder composed of plate-shaped particles from the viewpoint of controlling the Vvc of the magnetic layer surface and the width direction σ of the Vvc within the range described above. Further, from the viewpoint of controlling the Vvc of the magnetic layer surface and the width direction σ of the Vvc within the range described above, it was also clarified from the studies of the present inventor that a non-magnetic powder having an average plate diameter of 50 nm or more and 1,000 nm or less and an average plate thickness of 12 nm or less is more preferable as the non-magnetic powder of the back coating layer. For example, the average plate thickness can be 3 nm or more, 4 nm or more, or 5 nm or more, and can also be less than the value exemplified here. The non-magnetic powder composed of a plate-shaped particle (preferably, a non-magnetic powder having an average plate diameter of 50 nm or more and 1,000 nm or less and an average plate thickness of 12 nm or less) can be, for example, a metal oxide powder. Specific examples of the metal oxide include aluminum oxide, zirconium oxide, and cerium oxide. Aluminum oxide is also called alumina, zirconium oxide is also called zirconia, and cerium oxide is also called ceria. For example, by heating an aqueous solution of a metal hydroxide at a liquid temperature of 200° C. or higher to proceed with a hydrothermal synthesis reaction, plate-shaped metal oxide particles can be obtained. By adjusting various conditions for the particle preparation (for example, the concentration of the aqueous solution, the mixing time in a case of preparing the aqueous solution, the heating temperature, and the heating time), the plate diameter and the plate thickness of the plate-shaped particles can be controlled.

(144) The content (filling percentage) of the non-magnetic powder in the back coating layer is preferably in a range of 50 to 90 mass % and more preferably in a range of 60 to 90 mass % with respect to the total mass of the back coating layer.

(145) The composition for forming a back coating layer preferably contains a component (dispersing agent) capable of improving the dispersibility of the non-magnetic powder contained in this composition. The present inventor supposes that it is preferable to form the back coating layer by using a composition for forming a back coating layer containing such a dispersing agent from the viewpoint of reducing the influence of show-through and controlling the Vvc of the magnetic layer surface and the width direction σ of the Vvc within the range described above. As an example of such a dispersing agent, a compound having an ammonium salt structure of an alkyl ester anion represented by Formula 1 can be used. The “alkyl ester anion” can also be called an “alkyl carboxylate anion”.

(146) ##STR00001##

(147) In Formula 1, R represents an alkyl group having 7 or more carbon atoms or a fluorinated alkyl group having 7 or more carbon atoms, and Z.sup.+ represents an ammonium cation.

(148) In addition, from the viewpoint of improving the dispersibility of non-magnetic powder, in one aspect, two or more kinds of components capable of forming the compound having the salt

structure can be used in a case of preparing the composition for forming a back coating layer. Thereby, in a case where the composition for forming a back coating layer is prepared, at least a part of these components can form the compound having a salt structure.

(149) Unless otherwise noted, groups described below may have a substituent or may be unsubstituted. In addition, for a group having a substituent, the term “carbon atoms” means the number of carbon atoms not including the number of carbon atoms of the substituent, unless otherwise noted. In the present invention and the present specification, examples of the substituent include an alkyl group (for example, an alkyl group having 1 to 6 carbon atoms), a hydroxy group, an alkoxy group (for example, an alkoxy group having 1 to 6 carbon atoms), a halogen atom (for example, a fluorine atom, a chlorine atom, a bromine atom, or the like), a cyano group, an amino group, a nitro group, an acyl group, a carboxy group, a salt of a carboxy group, a sulfonic acid group, and a salt of a sulfonic acid group.

(150) Hereinafter, Formula 1 will be described in more detail.

(151) In Formula 1, R represents an alkyl group having 7 or more carbon atoms or a fluorinated alkyl group having 7 or more carbon atoms. The fluorinated alkyl group has a structure in which some or all of the hydrogen atoms constituting the alkyl group are substituted with fluorine atoms. The alkyl group or the fluorinated alkyl group represented by R may have a linear structure or a branched structure, may be a cyclic alkyl group or a fluorinated alkyl group, and is preferably a linear structure. The alkyl group or the fluorinated alkyl group represented by R may have a substituent, may be unsubstituted, and is preferably unsubstituted. The alkyl group represented by R can be represented by, for example, $C_{nH_{2n+1}}$. Here, n represents an integer of 7 or more. In addition, the fluorinated alkyl group represented by R may have a structure in which some or all of the hydrogen atoms constituting the alkyl group represented by, for example, C_nH_{2n+1} are substituted with fluorine atoms. The carbon number of the alkyl group or the fluorinated alkyl group represented by R is 7 or more, preferably 8 or more, more preferably 9 or more, still more preferably 10 or more, still more preferably 11 or more, still more preferably 12 or more, and still more preferably 13 or more. In addition, the carbon number of the alkyl group or the fluorinated alkyl group represented by R is preferably 20 or less, more preferably 19 or less, and still more preferably 18 or less.

(152) In Formula 1, Z^+ represents an ammonium cation. Specifically, the ammonium cation has the following structure. In the present invention and the present specification, an asterisk in the formula representing a part of a compound represents a bonding position between a structure of the part and an adjacent atom.

(153) ##STR00002##

(154) A nitrogen cation N^+ of the ammonium cation and an oxygen anion O^- in Formula 1 may form a salt crosslinking group to form the ammonium salt structure of the alkyl ester anion represented by Formula 1. The fact that the compound having the ammonium salt structure of the alkyl ester anion represented by Formula 1 is contained in the back coating layer can be confirmed by analyzing the magnetic tape by X-ray photoelectron spectroscopy (electron spectroscopy for chemical analysis (ESCA)), infrared spectroscopy (IR), or the like.

(155) In one aspect, the ammonium cation represented by Z^+ may be provided, for example, by a nitrogen atom of a nitrogen-containing polymer being a cation. The nitrogen-containing polymer means a polymer including a nitrogen atom. In the present invention and the present specification, the term “polymer” is used to encompass a homopolymer and a copolymer. The nitrogen atom may be included as an atom constituting a main chain of the polymer in one aspect, and may be included as an atom constituting a side chain of the polymer in one aspect.

(156) As one aspect of the nitrogen-containing polymer, polyalkyleneimine can be exemplified. Polyalkyleneimine is a ring-opening polymer of alkyleneimine and is a polymer having a plurality of repeating units represented by Formula 2.

(157) ##STR00003##

(158) The ammonium cation represented by Z.sup.+ in Formula 1 may be provided by a nitrogen atom N constituting a main chain in Formula 2 being a nitrogen cation N.sup.+. Then, the ammonium salt structure can be formed with the alkyl ester anion, for example, as follows.

(159) ##STR00004##

(160) Hereinafter, Formula 2 will be described in more detail.

(161) In Formula 2, R.sup.1 and R.sup.2 each independently represent a hydrogen atom or an alkyl group, and n1 represents an integer of 2 or more.

(162) Examples of the alkyl group represented by R.sup.1 or R.sup.2 include an alkyl group having 1 to 6 carbon atoms, preferably an alkyl group having 1 to 3 carbon atoms, more preferably a methyl group or an ethyl group, and still more preferably a methyl group. The alkyl group represented by R.sup.1 or R.sup.2 is preferably an unsubstituted alkyl group. The combination of R.sup.1 and R.sup.2 in Formula 2 may be a form in which one is a hydrogen atom and the other is an alkyl group, a form in which both are hydrogen atoms, and a form in which both are alkyl groups (the same or different alkyl groups), and the form in which both are hydrogen atoms is preferable. As the alkyleneimine that provides the polyalkyleneimine, a structure having the lowest number of carbon atoms constituting a ring is ethyleneimine, and the number of carbon atoms in a main chain of the alkyleneimine (ethyleneimine) obtained by the ring opening of the ethyleneimine is 2. Therefore, n1 in Formula 2 is 2 or more. n1 in Formula 2 may be, for example, 10 or less, 8 or less, 6 or less, or 4 or less. The polyalkyleneimine may be a homopolymer including only the same structure as the repeating structure represented by Formula 2, or may be a copolymer including two or more different structures as the repeating structure represented by Formula 2. A number-average molecular weight of polyalkyleneimine that can be used to form the compound having the ammonium salt structure of the alkyl ester anion represented by Formula 1 may be, for example, 200 or more, preferably 300 or more, and more preferably 400 or more. The number-average molecular weight of the polyalkyleneimine may be, for example, 10,000 or less, preferably 5,000 or less, and more preferably 2,000 or less.

(163) In the present invention and the present specification, the average molecular weight (weight-average molecular weight and number-average molecular weight) means a value measured by gel permeation chromatography (GPC) with standard polystyrene conversion. Unless otherwise noted, the average molecular weight shown in the section of Examples described below is a value (polystyrene conversion value) obtained by standard polystyrene conversion of values measured under the following measurement conditions using GPC. GPC device: HLC-8220 (manufactured by Tosoh Corporation) Guard column: TSKguardcolumn Super HZM-H Column: TSKgel Super HZ 2000, TSKgel Super HZ 4000, TSKgel Super HZ-M (manufactured by Tosoh Corporation, 4.6 mm (inner diameter)×15.0 cm, three columns connected in series) Eluent: Tetrahydrofuran (THF), containing stabilizer (2,6-di-t-butyl-4-methylphenol) Flow rate of eluent: 0.35 mL/min Column temperature: 40° C. Inlet temperature: 40° C. Refractive index (RI) measurement temperature: 40° C. Sample concentration: 0.3 mass % Sample injection amount: 10 µL

(164) As another aspect of the nitrogen-containing polymer, polyallylamine can be exemplified. Polyallylamine is a polymer of allylamine and is a polymer having a plurality of repeating units represented by Formula 3.

(165) ##STR00005##

(166) The ammonium cation represented by Z.sup.+ in Formula 1 may be provided by a nitrogen atom N constituting an amino group of a side chain in Formula 3 being a nitrogen cation N.sup.+. Then, the ammonium salt structure can be formed with the alkyl ester anion, for example, as follows.

(167) ##STR00006##

(168) A weight-average molecular weight of polyallylamine that can be used to form the compound having the ammonium salt structure of the alkyl ester anion represented by Formula 1 may be, for example, 200 or more, preferably 1,000 or more, and more preferably 1,500 or more. The weight-

average molecular weight of the polyallylamine may be, for example, 15,000 or less, preferably 10,000 or less, and more preferably 8,000 or less.

(169) The fact that a compound having a structure derived from polyalkyleneimine or polyallylamine as the compound having the ammonium salt structure of the alkyl ester anion represented by Formula 1 is contained in the back coating layer can be confirmed by analyzing the back coating layer surface by time-of-flight secondary ion mass spectrometry (TOF-SIMS) or the like.

(170) The compound having the ammonium salt structure of the alkyl ester anion represented by Formula 1 may be a salt of the nitrogen-containing polymer and one or more kinds of fatty acids selected from the group consisting of fatty acids having 7 or more carbon atoms and fluorinated fatty acids having 7 or more carbon atoms. The nitrogen-containing polymer forming a salt may be one or more kinds of nitrogen-containing polymers, and may be, for example, a nitrogen-containing polymer selected from the group consisting of polyalkyleneimine and polyallylamine. The fatty acids forming a salt may be one or more kinds of fatty acids selected from the group consisting of fatty acids having 7 or more carbon atoms and fluorinated fatty acids having 7 or more carbon atoms. The fluorinated fatty acid has a structure in which some or all of the hydrogen atoms constituting an alkyl group bonded to a carboxy group COOH in the fatty acid are substituted with fluorine atoms. For example, the salt forming reaction can easily proceed by mixing the nitrogen-containing polymer and the above fatty acids at a room temperature. A room temperature is, for example, about 20° C. to 25° C. In one aspect, one or more kinds of nitrogen-containing polymers and one or more kinds of the fatty acids are used as components of the composition for forming a back coating layer, and these are mixed in a step of preparing the composition for forming a back coating layer to allow the salt forming reaction to proceed. In addition, in one aspect, the composition for forming a back coating layer can be prepared by mixing one or more kinds of nitrogen-containing polymers and one or more kinds of the fatty acids to form a salt before preparation of the composition for forming a back coating layer, and then using the salt as a component of the composition for forming a back coating layer. In a case where the nitrogen-containing polymer and the fatty acids are mixed to form an ammonium salt of the alkyl ester anion represented by Formula 1, a nitrogen atom constituting the nitrogen-containing polymer may react with a carboxy group of the fatty acids to form the following structure, and a form including such a structure is also included in the compound.

(171) ##STR00007##

(172) Examples of the fatty acids include fatty acids having an alkyl group described above as R in Formula 1 and fluorinated fatty acids having a fluorinated alkyl group described above as R in Formula 1.

(173) A mixing ratio of the nitrogen-containing polymer used to form the compound having the ammonium salt structure of the alkyl ester anion represented by Formula 1 to the fatty acid is preferably 10:90 to 90:10, more preferably 20:80 to 85:15, and still more preferably 30:70 to 80:20 as a mass ratio of the nitrogen-containing polymer:the fatty acids. The compound having the ammonium salt structure of the alkyl ester anion represented by Formula 1 can be used, for example, in an amount of 1.0 to 20.0 parts by mass and is preferably used in an amount of 1.0 to 10.0 parts by mass with respect to 100.0 parts by mass of the non-magnetic powder, in a case where the composition for forming a back coating layer is prepared. In addition, for example, in a case where the composition for forming a back coating layer is prepared, 0.1 to 10.0 parts by mass of the nitrogen-containing polymer can be used, and 0.2 to 8.0 parts by mass of the nitrogen-containing polymer are preferably used, per 100.0 parts by mass of the non-magnetic powder. The above fatty acids can be used, for example, in an amount of 0.05 to 10.0 parts by mass and are preferably used in an amount of 0.1 to 5.5 parts by mass, per 100.0 parts by mass of the non-magnetic powder.

(174) Regarding the components that can be included in the back coating layer, the back coating

layer can include a binding agent and can also include an additive. Regarding the binding agent and additive in the back coating layer, a well-known technology for the back coating layer can be applied, and a well-known technology for the formulation of the magnetic layer and/or the non-magnetic layer can also be applied. For example, for the back coating layer, descriptions disclosed in paragraphs 0018 to 0020 of JP2006-331625A and page 4, line 65, to page 5, line 38, of U.S. Pat. No. 7,029,774B can be referred to.

(175) Various Thicknesses

(176) Regarding a thickness (total thickness) of the magnetic tape, it has been required to increase the recording capacity (increase the capacity) of the magnetic recording medium with the enormous increase in the amount of information in recent years. As means for increasing the capacity of a tape-shaped magnetic recording medium (that is, a magnetic tape), a thickness of the magnetic tape may be reduced to increase a length of the magnetic tape accommodated in one roll of a magnetic tape cartridge. From this point, the thickness (total thickness) of the magnetic tape is preferably 5.6 μm or less, more preferably 5.5 μm or less, still more preferably 5.4 μm or less, still more preferably 5.3 μm or less, still more preferably 5.2 μm or less, still more preferably 5.0 μm or less, and still more preferably 4.8 μm or less. In addition, from the viewpoint of ease of handling, the thickness of the magnetic tape is preferably 3.0 μm or more, more preferably 3.5 μm or more, and still more preferably 4.0 μm or more.

(177) The thickness (total thickness) of the magnetic tape can be measured by the following method.

(178) Ten samples (for example, 5 to 10 cm in length) are cut out from any part of the magnetic tape, and these samples are stacked to measure the thickness. A value (thickness per sample) obtained by dividing the measured thickness by 1/10 is set as the total thickness. The thickness measurement can be performed using a well-known measuring instrument capable of measuring a thickness on the order of 0.1 μm .

(179) A thickness of the non-magnetic support is preferably 3.0 to 5.0 μm .

(180) A thickness of the magnetic layer can be optimized according to a saturation magnetization amount of a magnetic head used, a head gap length, a band of a recording signal, and the like, and is generally 0.01 μm to 0.15 μm , and, from the viewpoint of high-density recording, the thickness is preferably 0.02 μm to 0.12 μm and more preferably 0.03 μm to 0.1 μm . The magnetic layer need only be at least a single layer, the magnetic layer may be separated into two or more layers having different magnetic properties, and a configuration of a well-known multilayered magnetic layer can be applied as the magnetic layer. A thickness of the magnetic layer in a case where the magnetic layer is separated into two or more layers is a total thickness of the layers.

(181) A thickness of the non-magnetic layer is, for example, 0.1 to 1.5 μm , preferably 0.1 to 1.0 μm , and more preferably 0.1 to 0.7 μm .

(182) A thickness of the back coating layer is preferably 0.9 μm or less and more preferably 0.1 to 0.7 μm .

(183) Various thicknesses such as the thickness of the magnetic layer and the like can be obtained by the following method.

(184) A cross section of the magnetic tape in a thickness direction is exposed by an ion beam, and then the exposed cross section observation is performed using a scanning electron microscope or a transmission electron microscope. Various thicknesses can be obtained as an arithmetic average of thicknesses obtained at two optional points in the cross section observation. Alternatively, the various thicknesses can be obtained as a designed thickness calculated according to manufacturing conditions.

(185) Manufacturing Step

(186) Preparation of Composition for Forming Each Layer

(187) A composition for forming the magnetic layer, the non-magnetic layer, or the back coating layer usually includes a solvent together with the various components described above. As a

solvent, one kind or two or more kinds of various solvents usually used for manufacturing a coating type magnetic recording medium can be used. A solvent content of a composition for forming each layer is not particularly limited. For the solvent, a description disclosed in a paragraph 0153 of JP2011-216149A can be referred to. The concentration of solid contents and the solvent composition of the composition for forming each layer need only be appropriately adjusted in accordance with the handling suitability of the composition, the coating conditions, and the thickness of each layer to be formed. A step of preparing a composition for forming the magnetic layer, the non-magnetic layer, or the back coating layer can usually include at least a kneading step, a dispersing step, and, as necessary, a mixing step provided before and after these steps. Each step may be divided into two or more stages. Various components used in the preparation of the composition for forming each layer may be added at the beginning or during any step. In addition, each component may be separately added in two or more steps. For example, a binding agent may be added separately in a kneading step, a dispersing step, and a mixing step for adjusting a viscosity after dispersion. In the manufacturing step of the magnetic tape, a well-known manufacturing technology in the related art can be used in a part of the steps. In the kneading step, an open kneader, a continuous kneader, a pressure kneader, or a kneader having a strong kneading force such as an extruder can be used. Details of the kneading step are disclosed in JP1989-106338A (JP-H1-106338A) and JP1989-79274A (JP-H1-79274A). As a disperser, various well-known dispersers using a shearing force, such as a beads mill, a ball mill, a sand mill, or a homomixer, can be used. Dispersion beads can be preferably used for the dispersion. Examples of the dispersion beads include ceramic beads and glass beads, and zirconia beads are preferable. Two or more kinds of beads may be used in combination. A bead diameter (particle size) and a bead filling rate of the dispersion beads are not particularly limited and need only be set depending on a powder to be dispersed. The dispersion time is not particularly limited. The present inventor supposes that it is preferable to improve the dispersibility of the non-magnetic powder in a composition for forming a back coating layer from the viewpoint of reducing the influence of show-through and controlling the V_{vc} of the magnetic layer surface and the width direction σ of the V_{vc} within the range described above. From the viewpoint of improving dispersibility of the non-magnetic powder in the composition for forming the back coating layer, it is preferable to increase the dispersion time in the preparation of the composition for forming the back coating layer. The composition for forming each layer may be filtered by a well-known method before performing the coating step. The filtering can be performed by using a filter, for example. As the filter used in the filtering, a filter having a pore diameter of 0.01 to 3 μm (for example, a filter made of glass fiber or a filter made of polypropylene) can be used, for example.

(188) Coating Step, Cooling Step, and Heating and Drying Step

(189) The magnetic layer can be formed by directly applying the composition for forming a magnetic layer onto the non-magnetic support or performing multilayer applying of the composition for forming a magnetic layer with the composition for forming a non-magnetic layer sequentially or simultaneously. For details of the coating for forming each layer, a description disclosed in a paragraph 0066 of JP2010-231843A can be referred to.

(190) As described above, in one aspect, the magnetic tape may have the non-magnetic layer between the non-magnetic support and the magnetic layer. Such a magnetic tape can be preferably manufactured by sequential multilayer coating. A manufacturing step of performing the sequential multilayer coating can be preferably performed as follows. The non-magnetic layer is formed through a coating step of applying a composition for forming a non-magnetic layer onto a non-magnetic support to form a coating layer, and a heating and drying step of drying the formed coating layer by a heat treatment. Then, the magnetic layer is formed through a coating step of applying a composition for forming a magnetic layer onto the formed non-magnetic layer to form a coating layer, and a heating and drying step of drying the formed coating layer by a heat treatment. In one aspect, it is also possible to form a coating layer by performing the coating step using the

composition for forming a non-magnetic layer, and to perform the cooling step of cooling the coating layer between the coating step and the heating and drying step in a non-magnetic layer forming step of the manufacturing method in which the sequential multilayer coating is performed. (191) Hereinafter, an example of the manufacturing step of the magnetic tape will be described with reference to FIG. 8. Note that the present invention is not limited to the following examples. (192) FIG. 8 is a schematic step diagram showing an example of a step of manufacturing a magnetic tape including a non-magnetic layer and a magnetic layer in this order on one surface of a non-magnetic support and a back coating layer on the other surface thereof. In the example shown in FIG. 8, an operation of feeding a non-magnetic support (elongated film) from a feeding part and winding the non-magnetic support around a winding part is continuously performed, and various treatments of coating, drying, and alignment are performed in each part or each zone shown in FIG. 8, and thus, it is possible to form a non-magnetic layer and a magnetic layer on one surface of the running non-magnetic support by sequential multilayer coating and to form a back coating layer on the other surface thereof. The example shown in FIG. 8 can be the same as the manufacturing step normally performed for manufacturing a coating type magnetic recording medium, except for including a cooling zone.

(193) The composition for forming a non-magnetic layer is applied onto the non-magnetic support fed from the feeding part in a first coating part (coating step of composition for forming a non-magnetic layer).

(194) After the coating step, a coating layer of the composition for forming a non-magnetic layer formed in the coating step is cooled in a cooling zone (cooling step). For example, the cooling step can be performed by allowing the non-magnetic support on which the coating layer is formed to pass through a cooling atmosphere. An atmosphere temperature of the cooling atmosphere may be preferably in a range of -10°C. to 0°C. , and more preferably in a range of -5°C. to 0°C. The time for performing the cooling step (for example, time while any part of the coating layer is delivered to and sent from the cooling zone (hereinafter, also referred to as a “staying time”)) is not particularly limited. In the cooling step, cooled air may blow to the surface of the coating layer.

(195) After the cooling zone, in a first heat treatment zone, the coating layer is heated after the cooling step to dry the coating layer (heating and drying step). The heating and drying treatment can be performed by allowing the non-magnetic support including the coating layer after the cooling step to pass through a heating atmosphere. An atmosphere temperature of the heating atmosphere here, and an atmosphere temperature of the heating atmosphere in the heating and drying step in a second heat treatment zone and the heating and drying step in a third heat treatment zone, which will be described below, are also referred to as a “drying temperature”. In one aspect, the drying temperature in each heat treatment zone is preferably 95°C. or higher, and more preferably 100°C. or higher. In addition, the drying temperature in each heat treatment zone may be, for example, 140°C. or lower or 130°C. or lower, and may be higher than the temperature described here. In addition, heated air may optionally blow to the surface of the coating layer.

(196) Next, in a second coating part, the composition for forming a magnetic layer is applied onto the non-magnetic layer formed by performing the heating and drying step in the first heat treatment zone (coating step of composition for forming a magnetic layer).

(197) After that, in an aspect of performing an alignment treatment, while the coating layer of the composition for forming a magnetic layer is in a wet state, an alignment treatment of the ferromagnetic powder in the coating layer is performed in an alignment zone. For the alignment treatment, various well-known technologies including a description disclosed in a paragraph 0067 of JP2010-231843A can be used. For example, a vertical alignment treatment can be performed by a well-known method such as a method using a polar opposing magnet. In the alignment zone, a drying speed of the coating layer can be controlled depending on a temperature of dry air and an air volume and/or a transportation speed of the magnetic tape in the alignment zone. Further, the coating layer may be preliminarily dried before the transportation to the alignment zone.

(198) The coating layer after the alignment treatment is subjected to the heating and drying step in the second heat treatment zone.

(199) Next, in a third coating part, a composition for forming a back coating layer is applied onto a surface of the non-magnetic support opposite to the surface on which the non-magnetic layer and the magnetic layer are formed, to form a coating layer (coating step of composition for forming a back coating layer). After that, in the third heat treatment zone, the coating layer is heated and dried.

(200) By the above step, it is possible to obtain the magnetic tape including the non-magnetic layer and the magnetic layer in this order on one surface of the non-magnetic support and the back coating layer on the other surface thereof.

(201) Other Steps

(202) In the manufacturing step of the magnetic tape, a calendering treatment is usually performed in order to improve the surface smoothness of the magnetic tape. Strengthening the calendering treatment conditions can contribute to reduction of the value of the width direction σ of the Vvc. Specific examples of strengthening the calendering treatment conditions include increasing a calender pressure, increasing a calender temperature, and decreasing a calender speed. Regarding the calendering treatment conditions, the calender pressure (linear pressure) is preferably 300 to 500 kN/m, and more preferably 310 to 350 kN/m, the calender temperature (surface temperature of a calender roll) is preferably 95° C. to 120° C., and more preferably 100° C. to 120° C., and the calender speed is preferably 50 to 75 m/min.

(203) For other various steps for manufacturing the magnetic tape, descriptions disclosed in paragraphs 0067 to 0070 of JP2010-231843A can be referred to.

(204) Through various steps, a long magnetic tape original roll can be obtained. The obtained magnetic tape original roll is cut (slit) by a well-known cutter, for example, to have a width of the magnetic tape to be accommodated in the magnetic tape cartridge. The width can be determined according to the standard, and is usually ½ inches. 1 inch=2.54 cm.

(205) A servo pattern is usually formed on the magnetic tape obtained by slitting.

(206) Formation of Servo Pattern

(207) The term “formation of servo pattern” can also be referred to as “recording of servo signal”. The formation of the servo pattern will be described below.

(208) The servo pattern is usually formed along a longitudinal direction of the magnetic tape. Examples of control (servo control) systems using a servo signal include a timing-based servo (TBS), an amplitude servo, and a frequency servo.

(209) As shown in a European computer manufacturers association (ECMA)-319 (June 2001), a magnetic tape conforming to a linear tape-open (LTO) standard (generally called “LTO tape”) employs a timing-based servo system. In this timing-based servo system, the servo pattern is formed by continuously arranging a plurality of pairs of non-parallel magnetic stripes (also referred to as “servo stripes”) in the longitudinal direction of the magnetic tape. In the present invention and the present specification, the term “timing-based servo pattern” refers to a servo pattern that enables head tracking in a timing-based servo system. As described above, the reason why the servo pattern is formed of a pair of non-parallel magnetic stripes is to indicate, to a servo signal reading element passing over the servo pattern, a passing position thereof. Specifically, the pair of magnetic stripes is formed such that an interval thereof continuously changes along a width direction of the magnetic tape, and the servo signal reading element reads the interval to thereby sense a relative position between the servo pattern and the servo signal reading element.

Information on this relative position enables tracking on a data track. Accordingly, a plurality of servo tracks are usually set on the servo pattern along the width direction of the magnetic tape.

(210) A servo band is formed of a servo pattern continuous in the longitudinal direction of the magnetic tape. A plurality of the servo bands are usually provided on the magnetic tape. For example, in an LTO tape, the number of the servo bands is five. Regions interposed between two

adjacent servo bands are data bands. The data band is formed of a plurality of data tracks and each data track corresponds to each servo track.

(211) Further, in one aspect, as shown in JP2004-318983A, information indicating a servo band number (referred to as “servo band identification (ID)” or “unique data band identification method (UDIM) information”) is embedded in each servo band. This servo band ID is recorded by shifting a specific one of the plurality of pairs of the servo stripes in the servo band so that positions thereof are relatively displaced in the longitudinal direction of the magnetic tape. Specifically, a way of shifting the specific one of the plurality of pairs of servo stripes is changed for each servo band. Accordingly, the recorded servo band ID is unique for each servo band, and thus, the servo band can be uniquely specified only by reading one servo band with a servo signal reading element.

(212) In a method of uniquely specifying the servo band, a staggered method as shown in ECMA-319 (June 2001) is used. In this staggered method, a group of pairs of non-parallel magnetic stripes (servo stripes) arranged continuously in plural in a longitudinal direction of the magnetic tape is recorded so as to be shifted in a longitudinal direction of the magnetic tape for each servo band. Since this combination of shifting methods between adjacent servo bands is unique throughout the magnetic tape, it is possible to uniquely specify a servo band in a case of reading a servo pattern with two servo signal reading elements.

(213) As shown in ECMA-319 (June 2001), information indicating a position of the magnetic tape in the longitudinal direction (also referred to as “longitudinal position (LPOS) information”) is usually embedded in each servo band. This LPOS information is also recorded by shifting the positions of the pair of servo stripes in the longitudinal direction of the magnetic tape, as the UDIM information. Note that, unlike the UDIM information, in this LPOS information, the same signal is recorded in each servo band.

(214) It is also possible to embed, in the servo band, the other information different from the above UDIM information and LPOS information. In this case, the embedded information may be different for each servo band as the UDIM information or may be common to all servo bands as the LPOS information.

(215) As a method of embedding information in the servo band, it is possible to employ a method other than the above. For example, a predetermined code may be recorded by thinning out a predetermined pair from the group of pairs of servo stripes.

(216) A head for forming a servo pattern is called a servo write head. The servo write head usually has a pair of gaps corresponding to the pair of magnetic stripes as many as the number of servo bands. Usually, a core and a coil are connected to each pair of gaps, and by supplying a current pulse to the coil, a magnetic field generated in the core can cause generation of a leakage magnetic field in the pair of gaps. In a case of forming the servo pattern, by inputting a current pulse while running the magnetic tape on the servo write head, the magnetic pattern corresponding to the pair of gaps is transferred to the magnetic tape to form the servo pattern. A width of each gap can be appropriately set according to a density of the servo pattern to be formed. The width of each gap can be set to, for example, 1 μm or less, 1 to 10 μm , 10 μm or more, and the like.

(217) Before the servo pattern is formed on the magnetic tape, the magnetic tape is usually subjected to a demagnetization (erasing) treatment. This erasing treatment can be performed by applying a uniform magnetic field to the magnetic tape using a direct current magnet or an alternating current magnet. The erasing treatment includes direct current (DC) erasing and alternating current (AC) erasing. AC erasing is performed by gradually decreasing an intensity of the magnetic field while reversing a direction of the magnetic field applied to the magnetic tape. On the other hand, DC erasing is performed by applying a unidirectional magnetic field to the magnetic tape. As the DC erasing, there are two additional methods. A first method is horizontal DC erasing of applying a unidirectional magnetic field along a longitudinal direction of the magnetic tape. A second method is vertical DC erasing of applying a unidirectional magnetic field along a thickness direction of the magnetic tape. The erasing treatment may be performed on the

entire magnetic tape or may be performed for each servo band of the magnetic tape.

(218) A direction of the magnetic field of the servo pattern to be formed is determined according to a direction of the erasing. For example, in a case where the horizontal DC erasing is performed to the magnetic tape, the servo pattern is formed so that the direction of the magnetic field is opposite to the direction of the erasing. Therefore, an output of a servo signal obtained by reading the servo pattern can be increased. As shown in JP2012-53940A, in a case where the magnetic pattern is transferred to, using the gap, a magnetic tape that has been subjected to the vertical DC erasing, a servo signal obtained by reading the formed servo pattern has a monopolar pulse shape. On the other hand, in a case where a magnetic pattern is transferred to, using the gap, a magnetic tape that has been subjected to horizontal DC erasing, a servo signal obtained by reading the formed servo pattern has a bipolar pulse shape.

(219) Vertical Squareness Ratio

(220) In one aspect, the vertical squareness ratio of the magnetic tape, may be, for example, 0.55 or more, and from the viewpoint of improving the electromagnetic conversion characteristics, the vertical squareness ratio is preferably 0.60 or more, and more preferably 0.65 or more. In principle, the upper limit of the squareness ratio is 1.00 or less. The vertical squareness ratio of the magnetic tape may be 1.00 or less, 0.95 or less, 0.90 or less, 0.85 or less, or 0.80 or less. From the viewpoint of improving the electromagnetic conversion characteristics, a large value of the vertical squareness ratio of the magnetic tape is preferable. The vertical squareness ratio of the magnetic tape can be controlled by a well-known method such as performing a vertical alignment treatment.

(221) In the present invention and the present specification, the term “vertical squareness ratio” refers to a squareness ratio measured in the vertical direction of the magnetic tape. The term “vertical direction” described regarding the squareness ratio refers to a direction orthogonal to the magnetic layer surface, and can also be referred to as a thickness direction. In the present invention and the present specification, the vertical squareness ratio is obtained by the following method.

(222) A sample piece having a size capable of being introduced into a vibrating sample magnetometer is cut out from the magnetic tape to be measured. For this sample piece, using a vibrating sample magnetometer, a magnetic field is applied in the vertical direction (direction orthogonal to the magnetic layer surface) of the sample piece at a maximum applied magnetic field of 3979 kA/m, a measurement temperature of 296 K, and a magnetic field sweeping speed of 8.3 kA/m/sec, and the magnetization strength of the sample piece with respect to the applied magnetic field is measured. The measured value of the magnetization strength is obtained as a value after demagnetic field correction and as a value obtained by subtracting the magnetization of a sample probe of the vibrating sample magnetometer as a background noise. Assuming that the magnetization strength at the maximum applied magnetic field is M_s and the magnetization strength at zero applied magnetic field is M_r , a squareness ratio SQ is a value calculated as $SQ = M_r / M_s$. The measurement temperature refers to a temperature of the sample piece, and, by setting an atmosphere temperature around the sample piece to a measurement temperature, the temperature of the sample piece can be set to a measurement temperature by establishing a temperature equilibrium.

(223) Magnetic Tape Cartridge

(224) Another aspect of the present invention relates to a magnetic tape cartridge comprising the magnetic tape described above.

(225) The details of the magnetic tape included in the above magnetic tape cartridge are as described above. The magnetic tape cartridge can be mounted on the magnetic tape apparatus comprising the magnetic head and used for recording and/or reproducing data.

(226) In the magnetic tape cartridge, generally, the magnetic tape is accommodated inside a cartridge body in a state of being wound around a reel. The reel is rotatably provided inside the cartridge body. As the magnetic tape cartridge, a single reel type magnetic tape cartridge having one reel inside the cartridge body and a dual reel type magnetic tape cartridge having two reels

inside the cartridge body are widely used. In a case where the single reel type magnetic tape cartridge is mounted on a magnetic tape apparatus for recording and/or reproducing data on the magnetic tape, the magnetic tape is pulled out of the magnetic tape cartridge to be wound around the reel on the magnetic tape apparatus side. A magnetic head is disposed on a magnetic tape transportation path from the magnetic tape cartridge to a winding reel. Sending and winding of the magnetic tape are performed between a reel (supply reel) on the magnetic tape cartridge side and a reel (winding reel) on the magnetic tape apparatus side. For example, during this time, data is recorded and/or reproduced as the magnetic head and the magnetic layer surface of the magnetic tape come into contact with each other to be slid on each other. With respect to this, in the dual reel type magnetic tape cartridge, both reels of the supply reel and the winding reel are provided in the magnetic tape cartridge.

(227) In one aspect, the magnetic tape cartridge may include a cartridge memory. The cartridge memory can be, for example, a non-volatile memory, and, in one aspect, head tilt angle adjustment information is already recorded or the head tilt angle adjustment information is recorded. The head tilt angle adjustment information is information for adjusting the head tilt angle during running of the magnetic tape in the magnetic tape apparatus. For example, as the head tilt angle adjustment information, the value of the servo band interval at each position in the longitudinal direction of the magnetic tape during data recording can be recorded. For example, in a case of reproducing the data recorded on the magnetic tape, the value of the servo band interval can be measured during reproduction, and the head tilt angle can be changed by a control device of the magnetic tape apparatus such that the absolute value of the difference from the servo band interval during recording at the same longitudinal position recorded in the cartridge memory approaches zero. The head tilt angle may be, for example, the angle θ described above.

(228) The magnetic tape and the magnetic tape cartridge can be suitably used in a magnetic tape apparatus (in other words, a magnetic recording and reproducing system) that records and/or reproduces data at different head tilt angles. In such a magnetic tape apparatus, in one aspect, it is possible to record and/or reproduce the data by changing the head tilt angle during running of the magnetic tape. For example, the head tilt angle can be changed according to the information on the dimensions in the width direction of the magnetic tape acquired during running of the magnetic tape. There is also a use form, for example, in which the head tilt angle in one recording and/or reproduction and the head tilt angle in subsequent recording and/or reproduction are changed, and then the head tilt angle is fixed without change during running of the magnetic tape for each recording and/or reproduction.

(229) Magnetic Tape Apparatus

(230) Still another aspect of the present invention relates to a magnetic tape apparatus including the magnetic tape described above. In the magnetic tape apparatus, recording of data on the magnetic tape and/or reproduction of data recorded on the magnetic tape can be performed, for example, as the magnetic layer surface of the magnetic tape and the magnetic head come into contact with each other to be slid on each other.

(231) In one aspect, the magnetic tape is treated as a removable medium (so-called replaceable medium), and a magnetic tape cartridge accommodating the magnetic tape therein is inserted into the magnetic tape apparatus and taken out. In another aspect, the magnetic tape is not treated as a replaceable medium, the magnetic tape is wound around the reel of the magnetic tape apparatus comprising a magnetic head, and the magnetic tape is accommodated in the magnetic tape apparatus.

(232) In the present invention and the present specification, the term “magnetic tape apparatus” means an apparatus capable of performing at least one of the recording of data on the magnetic tape or the reproduction of data recorded on the magnetic tape. Such an apparatus is generally called a drive.

(233) Magnetic Head

(234) The magnetic tape apparatus may include a magnetic head. The configuration of the magnetic head and the angle θ , which is the head tilt angle, are as described above with reference to FIGS. 5 to 7. In a case where the magnetic head includes a reproducing element, a magnetoresistive (MR) element capable of sensitively reading information recorded on the magnetic tape is preferable as the reproducing element. As the MR element, various well-known MR elements (for example, a giant magnetoresistive (GMR) element and a tunnel magnetoresistive (TMR) element) can be used. Hereinafter, a magnetic head that records data and/or reproduces recorded data will also be referred to as a “recording and reproducing head”. An element for recording data (recording element) and an element for reproducing data (reproducing element) are collectively referred to as a “magnetic head element”.

(235) In a case of recording data and/or reproducing recorded data, first, tracking using the servo signal can be performed. That is, by causing the servo signal reading element to follow a predetermined servo track, the magnetic head element can be controlled to pass on the target data track. Displacement of the data track is performed by changing a servo track read by the servo signal reading element in a tape width direction.

(236) The recording and reproducing head can also perform recording and/or reproduction with respect to other data bands. In this case, the servo signal reading element need only be displaced to a predetermined servo band using the above described UDIM information to start tracking for the servo band.

(237) FIG. 9 shows an arrangement example of data bands and servo bands. In FIG. 9, a plurality of servo bands 1 are arranged to be interposed between guide bands 3 in a magnetic layer of a magnetic tape MT. A plurality of regions 2 interposed between two servo bands are data bands. The servo pattern is a magnetization region, and is formed by magnetizing a specific region of the magnetic layer by the servo write head. A region magnetized by the servo write head (a position where the servo pattern is formed) is determined by the standard. For example, in an LTO Ultrium format tape which is based on a local standard, a plurality of servo patterns inclined with respect to a tape width direction as shown in FIG. 10 are formed on a servo band, in a case of manufacturing a magnetic tape. Specifically, in FIG. 10, a servo frame SF on the servo band 1 is composed of a servo sub-frame 1 (SSF1) and a servo sub-frame 2 (SSF2). The servo sub-frame 1 is composed of an A burst (in FIG. 10, reference numeral A) and a B burst (in FIG. 10, reference numeral B). The A burst is composed of servo patterns A1 to A5 and the B burst is composed of servo patterns B1 to B5. Meanwhile, the servo sub-frame 2 is composed of a C burst (in FIG. 10, reference numeral C) and a D burst (in FIG. 10, reference numeral D). The C burst is composed of servo patterns C1 to C4 and the D burst is composed of servo patterns D1 to D4. Such 18 servo patterns are arranged in the sub-frames in an array of 5, 5, 4, 4, as the sets of 5 servo patterns and 4 servo patterns, and are used for identifying the servo frames. FIG. 10 shows one servo frame for description. Note that, in practice, a plurality of the servo frames are arranged in the running direction in each servo band in the magnetic layer of the magnetic tape on which the head tracking of the timing-based servo system is performed. In FIG. 10, the arrow indicates the running direction of the magnetic tape. For example, an LTO Ultrium format tape usually has 5000 or more servo frames per 1 m of tape length in each servo band of the magnetic layer.

(238) In one aspect, in the magnetic tape apparatus, the head tilt angle can be changed during running of the magnetic tape in the magnetic tape apparatus. The head tilt angle is, for example, an angle θ formed by the axis of the element array with respect to the width direction of the magnetic tape. The angle θ is as described above. For example, by providing the recording and reproducing head unit of the magnetic head with an angle adjustment unit for adjusting the angle of the module of the magnetic head, the angle θ can be variably adjusted during running of the magnetic tape. Such an angle adjustment unit can include, for example, a rotation mechanism for rotating the module. A well-known technology can be applied to the angle adjustment unit.

(239) Regarding the head tilt angle during running of the magnetic tape, in a case where the

magnetic head includes a plurality of modules, it is possible to specify the angle θ described with reference to FIGS. 5 to 7 for randomly selected modules. $\theta_{\text{sub.initial}}$, which is an angle θ at the start of running of the magnetic tape, can be set to 0° or more or more than 0° . This is preferable in terms of adjustment ability to adjust the effective distance between the servo signal reading elements in response to the dimension change in the width direction of the magnetic tape, because the larger $\theta_{\text{sub.initial}}$, the larger the amount of change in the effective distance between the servo signal reading elements with respect to the amount of change in the angle θ . From this point, $\theta_{\text{sub.initial}}$ is preferably 1° or more, more preferably 5° or more, and still more preferably 10° or more. On the other hand, for an angle (generally referred to as “wrap angle”) formed between the magnetic layer surface and the contact surface of the magnetic head in a case where the magnetic tape runs and comes into contact with the magnetic head, it is effective to keep the deviation with respect to the tape width direction small in order to improve the uniformity in the tape width direction of the friction generated by the contact between the magnetic head and the magnetic tape during running of the magnetic tape. In addition, it is desirable to increase the uniformity of the friction in the tape width direction from the viewpoint of the position followability and the running stability of the magnetic head. From the viewpoint of reducing the deviation of the wrap angle in the tape width direction, $\theta_{\text{sub.initial}}$ is preferably 45° or less, more preferably 40° or less, and still more preferably 35° or less.

(240) Regarding the change in angle θ during running of the magnetic tape, in a case where the angle θ of the magnetic head changes from $\theta_{\text{sub.initial}}$ at the start of running while the magnetic tape runs in the magnetic tape apparatus for the recording of data on the magnetic tape and/or for the reproduction of data recorded on the magnetic tape, the maximum change amount $\Delta\theta$ of the angle θ during the running of the magnetic tape is the larger value between $\Delta\theta_{\text{sub.max}}$ and $\Delta\theta_{\text{sub.min}}$ calculated by the following equation. The maximum value of the angle θ during running of the magnetic tape is $\theta_{\text{sub.max}}$, and the minimum value is $\theta_{\text{sub.min}}$. Note that “max” is an abbreviation for maximum, and “min” is an abbreviation for minimum.

$$(241) \quad \theta_{\text{sub.max}} = \theta_{\text{sub.max}} - \theta_{\text{sub.initial}} \quad \theta_{\text{sub.min}} = \theta_{\text{sub.initial}} - \theta_{\text{sub.min}}$$

(242) In one aspect, $\Delta\theta$ may be more than 0.000° , and, from the viewpoint of the adjustment ability to adjust the effective distance between the servo signal reading elements in response to the dimension change in the width direction of the magnetic tape, $\Delta\theta$ is preferably 0.001° or more and more preferably 0.010° or more. From the viewpoint of easiness of ensuring synchronization of the recorded data and/or the reproduced data between a plurality of magnetic head elements during the recording and/or reproduction of the data, $\Delta\theta$ is preferably 1.000° or less, more preferably 0.900° or less, still more preferably 0.800° or less, still more preferably 0.700° or less, and still more preferably 0.600° or less.

(243) In the example shown in FIG. 6 and FIG. 7, the axis of the element array is tilted in the running direction of the magnetic tape. Note that the present invention is not limited to such an example. In the above-described magnetic tape apparatus, an embodiment in which the axis of the element array is tilted in a direction opposite to the running direction of the magnetic tape is also included in the present invention.

(244) $\theta_{\text{sub.initial}}$, which is the head tilt angle at the start of running of the magnetic tape, can be set by a control device of the magnetic tape apparatus or the like.

(245) Regarding the head tilt angle during running of the magnetic tape, FIG. 11 is an explanatory diagram of a measuring method of the angle θ during running of the magnetic tape. The angle θ during running of the magnetic tape can be obtained, for example, by the following method. In a case where the angle θ during running of the magnetic tape is obtained by the following method, the angle θ is changed in a range of 0° to 90° during running of the magnetic tape. That is, in a case where the axis of the element array is tilted in the running direction of the magnetic tape at the start of running of the magnetic tape, the element array is not tilted such that the axis of the element array is tilted in the direction opposite to the running direction of the magnetic tape at the start of

running of the magnetic tape, during running of magnetic tape, and in a case where the axis of the element array is tilted in the direction opposite to the running direction of the magnetic tape at the start of running of the magnetic tape, the element array is not tilted such that the axis of the element array is tilted in the running direction of the magnetic tape at the start of running of the magnetic tape, during running of the magnetic tape.

(246) A phase difference (that is, a time difference) ΔT between the reproduction signals of the pair of servo signal reading elements **1** and **2** is measured. The measurement of ΔT can be performed by a measurement unit provided in the magnetic tape apparatus. A configuration of such a measurement unit is well-known. The distance L between the central portion of the servo signal reading element **1** and the central portion of the servo signal reading element **2** can be measured by an optical microscope or the like. In a case where the running speed of the magnetic tape is a speed v , the distance between the central portions of the two servo signal reading elements in the running direction of the magnetic tape is $L \sin \theta$, and a relationship of $L \sin \theta = v \times \Delta T$ is established. Therefore, the angle θ during running of the magnetic tape can be calculated by the equation " $\theta = \arcsin(v\Delta T/L)$ ". The right figure of FIG. **11** shows an example in which the axis of the element array is tilted in the running direction of the magnetic tape. In this example, the phase difference (that is, the time difference) ΔT of the phase of the reproduction signal of the servo signal reading element **2** with respect to the phase of the reproduction signal of the servo signal reading element **1** is measured. In a case where the axis of the element array is tilted in the direction opposite to the running direction of the magnetic tape, θ can be obtained by the above-described method except for a point where ΔT is measured as the phase difference (that is, the time difference) of the phase of the reproduction signal of the servo signal reading element **1** with respect to the phase of the reproduction signal of the servo signal reading element **2**.

(247) A pitch suitable for a measurement pitch of the angle θ , that is, a measurement interval of the angle θ in the tape longitudinal direction can be selected according to a frequency of the tape width deformation in the tape longitudinal direction. As an example, the measurement pitch can be set to, for example, 250 μm .

(248) Configuration of Magnetic Tape Apparatus

(249) A magnetic tape apparatus **10** shown in FIG. **12** controls a recording and reproducing head unit **12** in accordance with an instruction from a control device **11**, and records and reproduces data on a magnetic tape MT.

(250) The magnetic tape apparatus **10** has a configuration capable of detecting and adjusting the tension applied in the longitudinal direction of the magnetic tape from spindle motors **17A** and **17B** for controlling rotation of a magnetic tape cartridge reel and a winding reel and driving devices **18A** and **18B** thereof.

(251) The magnetic tape apparatus **10** has a configuration capable of loading a magnetic tape cartridge **13**.

(252) The magnetic tape apparatus **10** has a cartridge memory reading and writing device **14** capable of reading and writing a cartridge memory **131** in the magnetic tape cartridge **13**.

(253) From the magnetic tape cartridge **13** mounted on the magnetic tape apparatus **10**, an end part or a leader pin of the magnetic tape MT is pulled out by an automatic loading mechanism or a manual operation, and the magnetic layer surface of the magnetic tape MT passes on the recording and reproducing head through guide rollers **15A** and **15B** in a direction contacting with a recording and reproducing head surface of the recording and reproducing head unit **12**, and thus the magnetic tape MT is wound around a winding reel **16**.

(254) The rotation and torque of the spindle motor **17A** and the spindle motor **17B** are controlled by a signal from the control device **11**, and the magnetic tape MT is run at any speed and tension. A servo pattern formed in advance on the magnetic tape can be used for a control of the tape speed and a control of the head tilt angle. In order to detect the tension, a tension detecting mechanism may be provided between the magnetic tape cartridge **13** and the winding reel **16**. The tension may

be controlled by using the guide rollers **15A** and **15B** in addition to the control by the spindle motors **17A** and **17B**.

(255) The cartridge memory reading and writing device **14** is configured to be capable of reading out and writing information in the cartridge memory **131** in response to an instruction from the control device **11**. As a communication method between the cartridge memory reading and writing device **14** and the cartridge memory **131**, for example, an international organization for standardization (ISO) 14443 method can be employed.

(256) The control device **11** includes, for example, a controller, a storage unit, a communication unit, and the like.

(257) The recording and reproducing head unit **12** includes, for example, a recording and reproducing head, a servo tracking actuator that adjusts a position of the recording and reproducing head in the track width direction, a recording and reproducing amplifier **19**, a connector cable for connection with the control device **11**, and the like. The recording and reproducing head includes, for example, a recording element for recording data on the magnetic tape, a reproducing element for reproducing data on the magnetic tape, and a servo signal reading element for reading a servo signal recorded on the magnetic tape. For example, one or more recording elements, reproducing elements, and servo signal reading elements are mounted in one magnetic head. Alternatively, each element may be separately provided in a plurality of magnetic heads according to the running direction of the magnetic tape.

(258) The recording and reproducing head unit **12** is configured to be capable of recording data on the magnetic tape MT in response to an instruction from the control device **11**. In addition, the recording and reproducing head unit **12** is configured to be capable of reproducing the data recorded on the magnetic tape MT in response to an instruction from the control device **11**.

(259) The control device **11** has a mechanism for obtaining the running position of the magnetic tape from the servo signal read from the servo band in a case where the magnetic tape MT is run, and controlling the servo tracking actuator such that the recording element and/or the reproducing element is located at a target running position (track position). The track position is controlled by feedback control, for example. The control device **11** has a mechanism for obtaining a servo band interval from servo signals read from two adjacent servo bands in a case where the magnetic tape MT is run. The control device **11** can store the obtained information on the servo band interval in the storage unit inside the control device **11**, the cartridge memory **131**, an external connection device, or the like. In addition, the control device **11** can change the head tilt angle according to the dimension information in the width direction of the magnetic tape during running. Accordingly, the effective distance between the servo signal reading elements can be made to approximate to or match with the interval between the servo bands. The dimension information can be acquired by using a servo pattern formed in advance on the magnetic tape. For example, in this way, during running of the magnetic tape in the magnetic tape apparatus, the angle θ formed by the axis of the element array with respect to the width direction of the magnetic tape can be changed according to the dimension information in the width direction of the magnetic tape acquired during running. The head tilt angle can be adjusted, for example, by feedback control. For example, the adjustment of the head tilt angle can also be performed by the method disclosed in JP2016-524774A or US2019/0164573A1.

EXAMPLES

(260) Hereinafter, the present invention will be described based on Examples. Note that the present invention is not limited to the embodiments shown in Examples. “Parts” and “%” in the following description mean “parts by mass” and “mass %”, respectively. The steps and evaluations in the following description were performed in an environment of a temperature of 23° C.±1° C., unless otherwise noted. “eq” in the following description is an equivalent and is a unit that cannot be converted into an SI unit.

(261) Ferromagnetic Powder

(262) In Table 1, “BaFe” is a hexagonal barium ferrite powder (coercivity H_c : 196 kA/m, average particle size (average plate diameter) of 24 nm).

(263) “SrFe1” shown in Table 1 is a hexagonal strontium ferrite powder manufactured by the following method.

(264) 1707 g of SrCO_3 , 687 g of H_3BO_3 , 1120 g of Fe_2O_3 , 45 g of Al(OH)_3 , 24 g of BaCO_3 , 13 g of CaCO_3 , and 235 g of Nd_2O_3 were weighed and mixed by a mixer to obtain a raw material mixture.

(265) The obtained raw material mixture was melted in a platinum crucible at a melting temperature of 1390°C ., and a hot water outlet provided at a bottom of the platinum crucible was heated while stirring a melt, and the melt was discharged in a rod shape at about 6 g/sec. Hot water was rolled and quenched by a pair of water-cooling rollers to manufacture an amorphous body.

(266) 280 g of the manufactured amorphous body was charged into an electric furnace, was heated to 635°C . (crystallization temperature) at a temperature rising rate of $3.5^\circ\text{C}/\text{min}$, and was kept at the same temperature for 5 hours to precipitate (crystallize) hexagonal strontium ferrite particles.

(267) Next, a crystallized product obtained above including hexagonal strontium ferrite particles was coarsely pulverized in a mortar, and 1000 g of zirconia beads having a particle diameter of 1 mm and 800 ml of an acetic acid aqueous solution of 1% concentration were added to a glass bottle containing the product, to be dispersed by a paint shaker for 3 hours. Thereafter, the obtained dispersion liquid was separated from the beads, to be put in a stainless beaker. The dispersion liquid was statically left at a liquid temperature of 100°C . for 3 hours and subjected to a dissolving treatment of a glass component, and then the crystallized product was sedimented by a centrifugal separator to be washed by repeatedly performing decantation and was dried in a heating furnace at an in-furnace temperature of 110°C . for 6 hours to obtain a hexagonal strontium ferrite powder.

(268) An average particle size of the hexagonal strontium ferrite powder obtained above was 18 nm, an activation volume was 902 nm^3 , an anisotropy constant K_u was $2.2 \times 10^4\text{ J/m}^3$, and a mass magnetization as was $49\text{ A m}^2/\text{kg}$.

(269) 12 mg of a sample powder was taken from the hexagonal strontium ferrite powder obtained above, elemental analysis of the filtrated solution obtained by partially dissolving this sample powder under dissolution conditions illustrated above was performed by an ICP analyzer, and a surface layer portion content of a neodymium atom was determined.

(270) Separately, 12 mg of a sample powder was taken from the hexagonal strontium ferrite powder obtained above, elemental analysis of the filtrated solution obtained by totally dissolving this sample powder under dissolution conditions illustrated above was performed by an ICP analyzer, and a bulk content of a neodymium atom was determined.

(271) A content (bulk content) of a neodymium atom with respect to 100 at % of an iron atom in the hexagonal strontium ferrite powder obtained above was 2.9 at %. A surface layer portion content of a neodymium atom was 8.0 at %. It was confirmed that a ratio between a surface layer portion content and a bulk content, that is, “surface layer portion content/bulk content” was 2.8, and a neodymium atom was unevenly distributed in a surface layer of a particle.

(272) The fact that the powder obtained above shows a crystal structure of hexagonal ferrite was confirmed by performing scanning with $\text{CuK}\alpha$ rays under conditions of a voltage of 45 kV and an intensity of 40 mA and measuring an X-ray diffraction pattern under the following conditions (X-ray diffraction analysis). The powder obtained above showed a crystal structure of hexagonal ferrite of a magnetoplumbite type (M type). A crystal phase detected by X-ray diffraction analysis was a single phase of a magnetoplumbite type. PANalytical X'Pert Pro diffractometer, PIXcel detector Soller slit of incident beam and diffracted beam: 0.017 radians Fixed angle of dispersion slit: $\frac{1}{4}$ degrees Mask: 10 mm Anti-scattering slit: $\frac{1}{4}$ degrees Measurement mode: continuous Measurement time per stage: 3 seconds Measurement speed: 0.017 degrees per second Measurement step: 0.05 degrees

(273) “SrFe2” shown in Table 1 is a hexagonal strontium ferrite powder manufactured by the

following method.

(274) 1725 g of SrCO_3 , 666 g of H_3BO_3 , 1332 g of Fe_2O_3 , 52 g of Al(OH)_3 , 34 g of CaCO_3 , and 141 g of BaCO_3 were weighed and mixed by a mixer to obtain a raw material mixture.

(275) The obtained raw material mixture was melted in a platinum crucible at a melting temperature of 1380°C ., and a hot water outlet provided at a bottom of the platinum crucible was heated while stirring a melt, and the melt was discharged in a rod shape at about 6 g/sec. Hot water was rolled and quenched by a pair of water-cooling rollers to manufacture an amorphous body.

(276) 280 g of the obtained amorphous body was charged into an electric furnace, was heated to 645°C . (crystallization temperature), and was held at the same temperature for 5 hours to precipitate (crystallize) hexagonal strontium ferrite particles.

(277) Next, a crystallized product obtained above including hexagonal strontium ferrite particles was coarsely pulverized in a mortar, and 1000 g of zirconia beads having a particle diameter of 1 mm and 800 ml of an acetic acid aqueous solution of 1% concentration were added to a glass bottle containing the product, to be dispersed by a paint shaker for 3 hours. Thereafter, the obtained dispersion liquid was separated from the beads, to be put in a stainless beaker. The dispersion liquid was statically left at a liquid temperature of 100°C . for 3 hours and subjected to a dissolving treatment of a glass component, and then the crystallized product was sedimented by a centrifugal separator to be washed by repeatedly performing decantation and was dried in a heating furnace at an in-furnace temperature of 110°C . for 6 hours to obtain a hexagonal strontium ferrite powder.

(278) An average particle size of the obtained hexagonal strontium ferrite powder was 19 nm, an activation volume was 1102 nm^3 , an anisotropy constant K_u was $2.0 \times 10^4\text{ J/m}^3$, and a mass magnetization as was $50\text{ A}\cdot\text{m}^2/\text{kg}$.

(279) In Table 1, “ ϵ -iron oxide” is an ϵ -iron oxide powder manufactured by the following method.

(280) 8.3 g of iron(III) nitrate nonahydrate, 1.3 g of gallium(III) nitrate octahydrate, 190 mg of cobalt(II) nitrate hexahydrate, 150 mg of titanium(IV) sulfate, and 1.5 g of polyvinylpyrrolidone (PVP) were dissolved in 90 g of pure water, and while the dissolved product was stirred using a magnetic stirrer, 4.0 g of an aqueous ammonia solution having a concentration of 25% was added to the dissolved product under a condition of an atmosphere temperature of 25°C . in an air atmosphere, and the dissolved product was stirred for 2 hours while maintaining a temperature condition of the atmosphere temperature of 25°C . A citric acid solution obtained by dissolving 1 g of citric acid in 9 g of pure water was added to the obtained solution, and the mixture was stirred for 1 hour. The powder sedimented after stirring was collected by centrifugal separation, was washed with pure water, and was dried in a heating furnace at an in-furnace temperature of 80°C .

(281) 800 g of pure water was added to the dried powder, and the powder was dispersed again in water to obtain dispersion liquid. The obtained dispersion liquid was heated to a liquid temperature of 50°C ., and 40 g of an aqueous ammonia solution having a concentration of 25% was dropwise added with stirring. After stirring for 1 hour while maintaining the temperature at 50°C ., 14 mL of tetraethoxysilane (TEOS) was added dropwise and was stirred for 24 hours. A powder sedimented by adding 50 g of ammonium sulfate to the obtained reaction solution was collected by centrifugal separation, was washed with pure water, and was dried in a heating furnace at an in-furnace temperature of 80°C . for 24 hours to obtain a ferromagnetic powder precursor.

(282) The obtained ferromagnetic powder precursor was loaded into a heating furnace at an in-furnace temperature of 1000°C . in an air atmosphere and was heat-treated for 4 hours.

(283) The heat-treated ferromagnetic powder precursor was put into an aqueous solution of 4 mol/L sodium hydroxide (NaOH), and the liquid temperature was maintained at 70°C . and was stirred for 24 hours, whereby a silicic acid compound as an impurity was removed from the heat-treated ferromagnetic powder precursor.

(284) Thereafter, the ferromagnetic powder from which the silicic acid compound was removed was collected by centrifugal separation, and was washed with pure water to obtain a ferromagnetic

powder.

(285) The composition of the obtained ferromagnetic powder that was checked by high-frequency inductively coupled plasma-optical emission spectrometry (ICP-OES) has Ga, Co, and a Ti substitution type ϵ -iron oxide (ϵ -Ga.sub.0.28Co.sub.0.05Ti.sub.0.05Fe.sub.1.62O.sub.3). In addition, X-ray diffraction analysis was performed under the same condition as that described above for SrFe1, and from a peak of an X-ray diffraction pattern, it was confirmed that the obtained ferromagnetic powder does not include α -phase and γ -phase crystal structures, and has a single-phase and ϵ -phase crystal structure (ϵ -iron oxide crystal structure).

(286) The obtained ϵ -iron oxide powder had an average particle size of 12 nm, an activation volume of 746 nm³, an anisotropy constant K_u of 1.2×10^5 J/m³, and a mass magnetization as of 16 A.Math.m.sup.2/kg.

(287) An activation volume and an anisotropy constant K_u of the above hexagonal strontium ferrite powder and ϵ -iron oxide powder are values obtained by the method described above using a vibrating sample magnetometer (manufactured by Toei Industry Co., Ltd.) for each ferromagnetic powder.

(288) In addition, a mass magnetization as is a value measured at a magnetic field intensity of 15 kOe using a vibrating sample magnetometer (manufactured by Toei Industry Co., Ltd.).

(289) Non-Magnetic Powder of Back Coating Layer

(290) In Examples and Comparative examples in which “Average plate diameter of non-magnetic powder” and “Average plate thickness of non-magnetic powder” are described in the column of “Back coating layer” in Table 1, the back coating layer was formed by using a non-magnetic powder composed of plate-shaped particles having an average plate diameter and an average plate thickness of values described in Table 1. The non-magnetic powder is a non-magnetic powder prepared by a method described below.

(291) The metal hydroxide and water were mixed in an airtight container (mixing time and molar concentration of the metal hydroxide aqueous solution: see Table 1). The mixed aqueous solution was heated for 10 days in an autoclave of which an internal temperature was set to 270° C. After heating, the generated particles were collected by centrifugal separation.

(292) In this way, the non-magnetic powder was prepared. An alumina powder was prepared by using aluminum hydroxide as the metal hydroxide, a zirconia powder was prepared by using zirconium hydroxide, and a ceria powder was prepared by using cerium hydroxide.

(293) In Table 1, “CS” described in “Type of non-magnetic powder” in the column of “Back coating layer” is an abbreviation of colloidal silica. The particle shape of the colloidal silica was spherical or elliptical.

(294) The “Average plate diameter of non-magnetic powder”, “Average plate thickness of non-magnetic powder”, and “Average particle size of non-magnetic powder” described in the column of “Back coating layer” in Table 1 are values obtained as described above by performing the measurement using a transmission electron microscope H-9000 manufactured by Hitachi, Ltd. as a transmission electron microscope and image analysis software KS-400 manufactured by Carl Zeiss as image analysis software.

Example 1

(295) (1) Formulation of Composition for Forming Magnetic Layer

(296) Magnetic Liquid Ferromagnetic powder (see Table 1): 100.0 parts SO.sub.3Na group-containing polyurethane resin: 14.0 parts Weight-average molecular weight: 70,000, SO.sub.3Na group: 0.4 meq/g Cyclohexanone: 150 parts Methyl ethyl ketone: 150 parts Abrasive Solution A Alumina abrasive (average particle size: 100 nm): 3.0 parts Sulfonic acid group-containing polyurethane resin: 0.3 parts Weight-average molecular weight: 70,000, SO.sub.3Na group: 0.3 meq/g Cyclohexanone: 26.7 parts Abrasive Solution B Diamond abrasive (average particle size: 100 nm): 1.0 part Sulfonic acid group-containing polyurethane resin: 0.1 parts Weight-average molecular weight: 70,000, SO.sub.3Na group: 0.3 meq/g Cyclohexanone: 26.7 parts Silica Sol

Colloidal silica (average particle size: 100 nm): 0.2 parts Methyl ethyl ketone: 1.4 parts Other Components Stearic acid: 2.0 parts Butyl stearate: 10.0 parts Polyisocyanate (CORONATE manufactured by Nippon Polyurethane Co., Ltd.): 2.5 parts Cyclohexanone: 200.0 parts Methyl ethyl ketone: 200.0 parts

(2) Formulation of Composition for Forming Non-Magnetic Layer α -Iron oxide powder (average particle volume: see Table 1): 100.0 parts Carbon black (average particle size: 20 nm, pH: see Table 1): 25.0 parts SO.sub.3Na group-containing polyurethane resin: 18.0 parts Weight-average molecular weight: 70,000, SO.sub.3Na group: 0.2 meq/g Stearic acid: 1.0 part Cyclohexanone: 300.0 parts Methyl ethyl ketone: 300.0 parts

(3) Formulation of Composition for Forming Back Coating Layer Non-magnetic powder (see Table 1 for details): 100.0 parts SO.sub.3Na group-containing polyurethane resin (SO.sub.3Na group: 70 eq/ton): 20.0 parts OSO.sub.3K group-containing vinyl chloride resin (OSO.sub.3K group: 70 eq/ton): 30.0 parts Polyethyleneimine (manufactured by Nippon Shokubai Co., Ltd., number-average molecular weight of 600): see Table 1 Stearic acid: see Table 1 Cyclohexanone: 140.0 parts Methyl ethyl ketone: 170.0 parts Butyl stearate: 2.0 parts Stearic acid amide: 0.1 parts

(4) Preparation of Composition for Forming Each Layer

(297) Regarding the composition for forming a magnetic layer, the above-mentioned components of the magnetic liquid were dispersed for 24 hours using a batch type vertical sand mill to prepare a magnetic liquid. As dispersion beads, zirconia beads having a bead diameter of 0.5 mm were used.

(298) Regarding the abrasive solution, the above-mentioned components of each of the abrasive solution A and the abrasive solution B were dispersed for 24 hours using a batch type ultrasonic device (20 kHz, 300 W) to obtain the abrasive solution A and the abrasive solution B.

(299) The magnetic liquid, the abrasive solution A, and the abrasive solution B were mixed with the above-mentioned silica sol and other components, and then subjected to a dispersion treatment for 30 minutes using a batch type ultrasonic device (20 kHz, 300 W). Thereafter, filtration was performed using a filter having a pore diameter of 0.5 μ m to prepare a composition for forming a magnetic layer.

(300) For the composition for forming a non-magnetic layer, the above-mentioned components were dispersed for 24 hours using a batch type vertical sand mill. As dispersion beads, zirconia beads having a bead diameter of 0.1 mm were used. The obtained dispersion liquid was filtered using a filter having a pore diameter of 0.5 μ m to prepare a composition for forming a non-magnetic layer.

(301) Regarding the composition for forming a back coating layer, the above-mentioned components were kneaded using a continuous kneader and then dispersed using a sand mill (dispersion time: see Table 1). 40.0 parts of polyisocyanate (CORONATE L manufactured by Nippon Polyurethane Industry Co., Ltd.) and 1000.0 parts of methyl ethyl ketone were added to the obtained dispersion liquid, and then the mixture was filtered using a filter having a pore diameter of 1 μ m to prepare a composition for forming a back coating layer.

(302) (5) Manufacturing of Magnetic Tape and Magnetic Tape Cartridge

(303) A magnetic tape was manufactured according to the manufacturing step shown in FIG. 8. The details are as follows.

(304) A polyethylene naphthalate support having a thickness of 3.7 μ m was fed from the feeding part, and the composition for forming a non-magnetic layer was applied onto one surface thereof so that the thickness after drying is 0.7 μ m in the first coating part, to form a coating layer. The cooling step was performed by allowing the formed coating layer to pass through the cooling zone in which the atmosphere temperature was adjusted to 0° C. while the coating layer is in a wet state (cooling zone staying time: 1 second), and then the heating and drying step was performed by allowing the coating layer to pass through the first heat treatment zone with the drying temperature (atmosphere temperature, the same applies hereinafter) of 105° C., to form a non-magnetic layer.

(305) After that, the composition for forming a magnetic layer prepared as described above was

applied onto the non-magnetic layer so that the thickness after drying is 0.1 μm in the second coating part, to form a coating layer. The vertical alignment treatment was performed in the alignment zone by applying a magnetic field having a magnetic field intensity of 0.5 T onto the surface of the coating layer of the composition for forming a magnetic layer in the vertical direction while the coating layer is in a wet state, and then the coating layer was dried in the second heat treatment zone with the drying temperature of 105° C.

(306) After that, in the third coating part, the composition for forming a back coating layer prepared as described above was applied onto the surface of the polyethylene naphthalate non-magnetic support opposite to the surface on which the non-magnetic layer and the magnetic layer are formed, so that the thickness after drying is 0.3 μm , to form a coating layer, and the formed coating layer was dried in the third heat treatment zone with the drying temperature of 105° C.

(307) After that, a calendering treatment (surface smoothing treatment) was performed under the calendering treatment conditions shown in Table 1 using a calender roll composed of only a metal roll.

(308) After that, a heat treatment was performed for 36 hours in an environment of an atmosphere temperature of 70° C. After the heat treatment, the resultant was slit to have ½ inches width to manufacture a magnetic tape.

(309) A servo signal was recorded on the magnetic layer of the manufactured magnetic tape by a commercially available servo writer in a state where the magnetic layer was demagnetized, to obtain a magnetic tape having a data band, a servo band, and a guide band in an arrangement according to a linear tape-open (LTO) Ultrium format and having a servo pattern (timing-based servo pattern) in an arrangement and a shape according to the LTO Ultrium format on the servo band. The servo pattern thus formed is a servo pattern according to the description in Japanese industrial standards (JIS) X6175:2006 and Standard ECMA-319 (June 2001). The total number of servo bands is 5, and the total number of data bands is 4.

(310) In this way, a magnetic tape (960 m in length) on which the servo signal was recorded was manufactured.

(311) It could be confirmed by the following method that the back coating layer of the magnetic tape includes a compound formed of polyethyleneimine and stearic acid and including the ammonium salt structure of the alkyl ester anion represented by Formula 1.

(312) A sample was cut out from the magnetic tape, and X-ray photoelectron spectroscopy analysis was performed on the back coating layer surface (measurement region: 300 μm ×700 μm) using an ESCA device. Specifically, the wide scanning measurement was performed by the ESCA device under the following measurement conditions. In measurement results, peaks were confirmed at a binding energy position of an ester anion and a binding energy position of an ammonium cation. Device: AXIS-ULTRA manufactured by Shimadzu Corporation Excited X-ray source: monochromatic Al-K α ray Scanning range: 0 to 1200 eV Pass energy: 160 eV Energy resolution: 1 eV/step Take-in time: 100 ms/step Accumulation number: 5

(313) In addition, a sample piece having a length of 3 cm was cut out from the magnetic tape, and the attenuated total reflection-fourier transform-infrared spectrometer (ATR-FT-IR) measurement (reflection method) was performed on the back coating layer surface. In measurement results, an absorption was confirmed at the wave number (1540 cm^{-1} or 1430 cm^{-1}) corresponding to an absorption of COO $^{-}$ and the wave number (2400 cm^{-1}) corresponding to an absorption of an ammonium cation.

Examples 2 to 16 and Comparative Examples 1 to 11

(314) A magnetic tape was obtained by the method described for Example 1 except that the items shown in Table 1 were changed as shown in Table 1.

(315) For each of Examples and Comparative Examples, four magnetic tapes having a length of 960 m were manufactured, and each of which was used for evaluation of (1) to (4) below.

Evaluation Method

(316) (1) Vvc of Magnetic Layer Surface and Width Direction σ of Vvc

(317) The following conditions were adopted as the measurement conditions of the AFM, and Vvc of the magnetic layer surface of the magnetic tape and the width direction σ of Vvc were obtained by the method described above. As the AFM data analysis software, AFM data analysis software (Nanoscope Analysis) provided by BRUKER was used.

(318) The region of the area of $40\text{ }\mu\text{m}\times 40\text{ }\mu\text{m}$ on the surface of the magnetic layer of the magnetic tape is measured with an AFM (Nanoscope 5 manufactured by BRUKER) in a peak fore-tapping mode. SCANASYST-AIR manufactured by BRUKER is used as a probe, a resolution is set to $512\text{ pixels}\times 512\text{ pixels}$, and a scan speed is set to a speed at which one screen ($512\text{ pixels}\times 512\text{ pixels}$) is measured in 512 seconds.

(319) (2) Non-Linear Component of Tape Width Deformation

(320) For each magnetic tape of Examples and Comparative Examples, a non-linear component of tape width deformation generated by storage for 10 days in an environment of a temperature of 60°C . and a relative humidity of 20% was measured by the method described above.

(321) (3) Total Thickness of Magnetic Tape (Tape Thickness)

(322) Ten tape samples (5 cm in length) were cut out from any part of each magnetic tape of Examples and Comparative Examples, and these tape samples were stacked to measure the thickness. The thickness was measured using a digital thickness gauge of Millimar 1240 compact amplifier and Millimar 1301 induction probe manufactured by MARH Inc. A value (thickness per tape sample) obtained by dividing the measured thickness by 1/10 was defined as the tape thickness. The tape thickness of any of the magnetic tapes was $4.8\text{ }\mu\text{m}$.

(323) Regarding the thicknesses of the magnetic layer, the non-magnetic layer, and the back coating layer of each of the magnetic tapes of Examples and Comparative Examples, cross-sectional observation was performed as described above, and it was confirmed that the respective thicknesses were the thicknesses described above.

(324) (4) Recording and Reproducing Performance

(325) The recording and reproducing performance of each magnetic tape of Examples and Comparative Examples was evaluated by the following method.

(326) As the magnetic head, a magnetic head comprising a reproducing module including an element array with 10 channels or more of a reproducing element having a reproducing element width of $0.2\text{ }\mu\text{m}$ or less between a pair of servo signal reading elements, and a recording module including an element array with 10 channels or more of a recording element having a recording element width, which is equal to or more than 1.5 times the reproducing element width, between a pair of servo signal reading elements was used. In the element array, an interval between two adjacent elements (that is, two adjacent reproducing elements and two adjacent recording elements) in the head width direction was $40\text{ }\mu\text{m}$ or more.

(327) The environment for recording and reproducing the data was such that a temperature is 20°C . to 25°C . and a relative humidity is 40% to 60%. Under such an environment, a magnetic tape apparatus in which the magnetic tape and the magnetic head were mounted to a tape transport system (reel tester) was placed for 24 hours or longer, and then data was recorded and reproduced. As a recording and reproducing amplifier to be mounted to the tape transport system of the magnetic tape apparatus, the recording and reproducing amplifier described above with respect to the measurement of the non-linear component of the tape width deformation was used. During the recording and reproduction of the data, the servo-following described above and the dynamic track position control (head tilt angle change) were executed. The recording and reproduction of the data were executed in detail as follows.

(328) A signal was recorded by the recording element while the magnetic tape was run at a constant speed of 5 m/sec. As a bit sequence to be recorded, a 255-bit pseudo random bit sequence (PRBS) generated according to a generating polynomial $x^8+x^4+x^3+x^2+x+1$ was used. The symbol “{circumflex over ()}” was used.

over ()}” represents a power. A linear recording density was set to 600 kbp. The unit “kbp” is a unit of the linear recording density (cannot be converted into an SI unit system). Single (shingled) recording of three or more tracks was performed such that a difference of $(PES1+PES2)/2$ between adjacent tracks was 1.5 times a reproduction track width.

(329) A magnetization pattern recorded on the magnetic tape was reproduced by the reproducing element immediately after recording (that is, the reproducing element with the same channel number), and the signal was amplified by the reproducing amplifier. The reproduction signal was decoded into a bit sequence based on phase lock loop (PLL) and auto gain control (AGC) processing, followed by a data dependent noise predictive maximum likelihood (DD-NPML) signal processing. Bit-by-bit comparison was made between the recorded bit sequence and the reproduced and decoded bit sequence, and in a case where the bits were different from each other, one-bit-error was counted. Data comparison was made over 10 Mbit, and a value obtained by dividing the accumulated error bit count by 10 Mbit was defined as a bit error rate. It was confirmed that the reproduction signal immediately after recording had a bit error rate of 1/1000 or less in all channels.

(330) Next, the magnetic tape was stored for 10 days in an environment of a temperature of 60° C. and a relative humidity of 20% in a state of being wound around a reel of the reel tester.

(331) After the above-mentioned storage, the magnetic tape was taken out from the storage environment and placed 24 hours or longer in an environment of a temperature of 20° C. to 25° C. and a relative humidity of 40% to 60% while being mounted to the same magnetic tape apparatus as the magnetic tape apparatus used before the storage, and then the data track recorded before the storage was reproduced (no recording was performed) in the same environment. In this case, the reproduction was performed only on the data tracks in which the data tracks were recorded on both sides. The bit error rates of all the channels were calculated, and the recording and reproducing performance was evaluated according to the following evaluation standard, with channels with a bit error rate of 1/100 or higher regarded as defective channels.

(332) Evaluation Standard

(333) A: A ratio of the number of defective channels to the total number of channels was less than 5%. B: A ratio of the number of defective channels to the total number of channels was 5% or more and less than 10%. C: A ratio of the number of defective channels to the total number of channels was 10% or more.

(334) The above results are shown in Table 1 (Tables 1-1 to 1-5).

(335) TABLE-US-00001

TABLE 1	Example 1	Example 2	Example 3	Example 4	Example 5
Magnetic Ferromagnetic	BaFe	BaFe	BaFe	BaFe	BaFe
layer powder	Type	Back coating			
Polyethyleneimine	0.2 parts	0.2 parts	1.0 part	1.0 part	0.2 parts
Stearic acid	0.4 parts	0.4 parts			
Type of non-magnetic powder	Alumina	Alumina	Alumina	Alumina	
Average particle size of non-magnetic powder	Average plate diameter of 1000 nm	500 nm			
500 nm	50 nm	1000 nm	non-magnetic powder	Average plate thickness of 12 nm	12 nm
7 nm	7 nm				
12 nm	non-magnetic powder	Molar concentration of metal	0.75 mol/L	0.50 mol/L	0.50 mol/L
0.10 mol/L	0.75 mol/L	hydroxide aqueous solution	Mixing time of metal	3 h	3 h
5 h	5 h	3 h	hydroxide		
Dispersion time	1 h	1 h	2 h	2 h	1 h
Non-magnetic α-Iron oxide powder	2.0 × 10.sup.-6	2.0 × 10.sup.-6	2.0 × 10.sup.-6	2.0 × 10.sup.-6	layer
Average particle volume	(μm.sup.3)	Carbon black	pH	9.0	9.0
9.0	9.0	9.0	9.0	9.0	Calendering
Temperature	100° C.	100° C.	100° C.	100° C.	
treatment	Linear pressure	320 kN/m	320 kN/m	320 kN/m	320 kN/m
condition	Speed	70 m/min	70 m/min	70 m/min	70 m/min
Vvc Unit: nm.sup.3	3.000	2.500	2.000	1.200	3.000
Width direction σ	Unit: nm.sup.3	0.120	0.120	0.100	0.090
0.120	of Vvc				
Non-linear Unit: nm	70	70	60	45	70
component of tape width deformation	Recording and	B	B	B	A
B reproducing performance	Example 6	Example 7	Example 8	Example 9	Example 10
Magnetic Ferromagnetic	BaFe	BaFe	BaFe	BaFe	BaFe
layer powder	Type	Back coating			
Polyethyleneimine	0.2 parts	1.0 part	1.0 part	0.2 parts	0.2 parts
Stearic acid	0.4 parts	2.0 parts	2.0 parts	0.4 parts	

0.4 parts Type of non-magnetic powder Zirconia Zirconia Ceria Ceria Average particle size of non-magnetic powder Average plate diameter of 500 nm 500 nm 50 nm 1000 nm 500 nm non-magnetic powder Average plate thickness of 12 nm 7 nm 7 nm 12 nm 12 nm non-magnetic powder Molar concentration of metal 0.50 mol/L 0.50 mol/L 0.10 mol/L 0.75 mol/L 0.50 mol/L hydroxide aqueous solution Mixing time of metal 3 h 5 h 5 h 3 h 3 h hydroxide and water Dispersion time 1 h 2 h 2 h 1 h 1 h Non-magnetic α -Iron oxide powder $2.0 \times 10^{sup.-6}$ $2.0 \times 10^{sup.-6}$ $2.0 \times 10^{sup.-6}$ $2.0 \times 10^{sup.-6}$ $2.0 \times 10^{sup.-6}$ layer Average particle volume ($\mu\text{m}^{sup.3}$) Carbon black pH 9.0 9.0 9.0 9.0 9.0 Calendering Temperature 100° C. 100° C. 100° C. 100° C. 100° C. treatment Linear pressure 320 kN/m 320 kN/m 320 kN/m 320 kN/m 320 kN/m condition Speed 70 m/min 70 m/min 70 m/min 70 m/min 70 m/min Vvc Unit: nm.^{sup.3} 2.500 2.000 1.200 3.000 2.500 Width direction σ Unit: nm.^{sup.3} 0.120 0.100 0.090 0.120 0.120 of Vvc Non-linear Unit: nm 70 60 45 70 70 component of tape width deformation Recording and B B A B B reproducing performance Example 11 Example 12 Example 13 Example 14 Example 15 Example 16 Magnetic Ferromagnetic BaFe BaFe BaFe SrFe1 SrFe2 ϵ -Iron oxide layer powder Type Back coating Polyethyleneimine 1.0 part 1.0 part 2.5 parts 0.2 parts 0.2 parts 0.2 parts layer Stearic acid 2.0 parts 2.0 parts 5.0 parts 0.4 parts 0.4 parts 0.4 parts Type of non- Ceria Ceria Alumina Alumina Alumina Alumina magnetic powder Average particle size of non- magnetic powder Average plate 500 nm 50 nm 50 nm 1000 nm 1000 nm 1000 nm diameter of non- magnetic powder Average plate 7 nm 7 nm 7 nm 12 nm 12 nm 12 nm thickness of non- magnetic powder Molar concentration 0.50 mol/L 0.10 mol/L 0.10 mol/L 0.75 mol/L 0.75 mol/L 0.75 mol/L of metal hydroxide aqueous solution Mixing time 5 h 5 h 5 h 3 h 3 h 3 h of metal hydroxide and water Dispersion time 2 h 2 h 3 h 1 h 1 h 1 h Non-magnetic α -Iron $2.0 \times 10^{sup.-6}$ $2.0 \times 10^{sup.-6}$ $2.0 \times 10^{sup.-6}$ $2.0 \times 10^{sup.-6}$ $2.0 \times 10^{sup.-6}$ layer oxide powder Average particle volume ($\mu\text{m}^{sup.3}$) Carbon black pH 9.0 9.0 8.0 9.0 9.0 9.0 Calendering Temperature 100° C. 100° C. 100° C. 100° C. 100° C. 100° C. treatment Linear pressure 320 kN/m 320 kN/m 320 kN/m 320 kN/m 320 kN/m 320 kN/m condition Speed 70 m/min 70 m/min 70 m/min 70 m/min 70 m/min 70 m/min Vvc Unit: nm.^{sup.3} 2.000 1.200 1.200 3.000 3.000 3.000 Width direction σ Unit: nm.^{sup.3} 0.100 0.090 0.060 0.120 0.120 0.120 of Vvc Non-linear Unit: nm 60 45 25 70 70 70 component of tape width deformation Recording and B A A B B B reproducing performance Comparative Comparative Comparative Comparative Example 1 Example 2 Example 3 Example 4 Example 5 Magnetic Ferromagnetic BaFe BaFe BaFe BaFe BaFe layer powder Type Back coating Polyethyleneimine 0 parts 0 parts 0 parts 0 parts 0 parts layer Stearic acid 0 parts 0 parts 0 parts 0 parts 0 parts Type of non-magnetic powder CS CS CS Alumina Alumina Average particle size of 100 nm 100 nm 50 nm non-magnetic powder Average plate diameter of 3000 nm 1000 nm non-magnetic powder Average plate thickness of 100 nm 15 nm non-magnetic powder Molar concentration of metal 1.00 mol/L 0.75 mol/L hydroxide aqueous solution Mixing time of metal 0.1 h 1 h hydroxide and water Dispersion time 0.5 h 1 h 2 h 1 h 1 h Non-magnetic α -Iron oxide powder $2.5 \times 10^{sup.-6}$ $2.5 \times 10^{sup.-6}$ $2.5 \times 10^{sup.-6}$ $2.5 \times 10^{sup.-6}$ $2.5 \times 10^{sup.-6}$ layer Average particle volume ($\mu\text{m}^{sup.3}$) Carbon black pH 11.0 11.0 11.0 11.0 11.0 Calendering Temperature 90° C. 90° C. 90° C. 90° C. 90° C. treatment Linear pressure 294 kN/mm 294 kN/mm 294 kN/mm 294 kN/mm 294 kN/mm condition Speed 80 m/min 80 m/min 80 m/min 80 m/min 80 m/min Vvc Unit: nm.^{sup.3} 4.500 4.000 3.500 3.000 2.800 Width direction σ Unit: nm.^{sup.3} 0.200 0.180 0.150 0.180 0.160 of Vvc Non-linear Unit: nm 200 150 130 125 120 component of tape width deformation Recording and C C C C C reproducing performance Comparative Comparative Comparative Comparative Comparative Comparative Comparative Example 6 Example 7 Example 8 Example 9 Example 10 Example 11 Magnetic Ferromagnetic BaFe BaFe BaFe BaFe BaFe BaFe layer powder Type Back Polyethyleneimine 0 parts 0 parts 0 parts 0 parts 2.0 parts 2.0 parts coating layer Stearic acid 0 parts 0 parts 0 parts 0 parts 5.0 parts 5.0 parts Type of non- Alumina CS CS CS Alumina Alumina magnetic powder Average particle 100 nm 100 nm 50 nm size of non- magnetic powder Average plate 40 nm 40 nm 40 nm diameter of non- magnetic powder Average plate 12 nm 7 nm 7 nm

thickness of non- magnetic powder Molar concentration 0.075 mol/L 0.075 mol/L 0.075 mol/L of metal hydroxide aqueous solution Mixing time of metal 3 h 5 h 5 h hydroxide and water Dispersion time 2 h 0.5 h 1 h 2 h 10 h 15 h Non-magnetic α -Iron oxide powder $2.5 \times 10^{sup.-6}$ $1.0 \times 10^{sup.-6}$ $1.0 \times 10^{sup.-6}$ $1.0 \times 10^{sup.-6}$ $1.0 \times 10^{sup.-6}$ $1.0 \times 10^{sup.-6}$ layer Average particle volume ($\mu\text{m}^{sup.3}$) Carbon black pH 11.0 8.0 8.0 8.0 8.0 8.0 Calendering Temperature 90° C. 90° C. 90° C. 90° C. 90° C. 90° C. treatment Linear pressure 294 kN/mm 294 kN/mm 294 kN/mm 294 kN/mm 294 kN/mm 294 kN/mm condition Speed 80 m/min 80 m/min 80 m/min 80 m/min 80 m/min 80 m/min Vvc Unit: nm^{sup.3} 2.500 4.500 4.000 3.500 1.000 0.900 Width direction σ Unit: nm^{sup.3} 0.150 0.120 0.120 0.100 0.012 0.010 of Vvc Non-linear Unit: nm 118 105 105 102 110 105 component of tape width deformation Recording and C C C C C C reproducing performance

(336) As shown in Table 1, the magnetic tapes of Examples showed superior recording and reproducing performance after being stored in an accelerated environment equivalent to long-term storage as compared with the magnetic tapes of Comparative Examples. From this result, it can be confirmed that the magnetic tapes of Examples contributed to the improvement of the operational stability of the drive (magnetic tape apparatus).

(337) A magnetic tape was manufactured by the method described in Example 1, except that the vertical alignment treatment was not performed in a case of manufacturing the magnetic tape.

(338) A sample piece was cut out from the magnetic tape. For this sample piece, a vertical squareness ratio obtained by the method described above using a TM-TRVSM5050-SMSL type manufactured by Tamakawa Co., Ltd. as a vibrating sample magnetometer was 0.55.

(339) The vertical squareness ratio similarly obtained for the sample piece cut out from the magnetic tape of Example 1 was 0.65.

(340) Each of the above two magnetic tapes was mounted to a reel tester of ½ inches, and electromagnetic conversion characteristics (signal-to-noise ratio (SNR)) were evaluated by the following method. As a result, for the magnetic tape of Example 1, a value of SNR higher by 4 dB was obtained as compared with the magnetic tape manufactured without the vertical alignment treatment.

(341) In an environment of a temperature of 23° C. and a relative humidity of 50%, a tension of 0.7 N (Newton) was applied in the longitudinal direction of the magnetic tape, and recording and reproduction were performed for 10 passes. A relative speed between the magnetic tape and the magnetic head was set to 6 m/sec, and recording was performed by using a metal-in-gap (MIG) head (a gap length of 0.15 μm and a track width of 1.0 μm) as a recording head and setting a recording current to an optimal recording current of each magnetic tape. Reproduction was performed by using a giant-magnetoresistive (GMR) head (an element thickness of 15 nm, a shield interval of 0.1 μm , and a reproducing element width of 0.8 μm) as a reproducing head. The head tilt angle was set to 0°. A signal having a linear recording density of 300 kfc/i was recorded, and measurement regarding a reproduction signal was performed with a spectrum analyzer manufactured by Shibasoku Co., Ltd. The unit kfc/i is a unit of a linear recording density (cannot be converted into an SI unit system). As the signal, a portion where the signal was sufficiently stable after start of the running of the magnetic tape was used.

(342) One aspect of the present invention is useful in various data storage technical fields.

Claims

1. A magnetic tape comprising: a non-magnetic support; and a magnetic layer containing a ferromagnetic powder, wherein a core void volume Vvc specified in ISO 25178 on a surface of the magnetic layer is 1.200 nm^{sup.3} or more and 3.000 nm^{sup.3} or less, and standard deviation σ of the Vvc in a width direction of the surface of the magnetic layer is 0.120 nm^{sup.3} or less.
2. The magnetic tape according to claim 1, wherein the standard deviation σ of the Vvc is 0.095 nm^{sup.3} or less.

3. The magnetic tape according to claim 1, further comprising: a non-magnetic layer containing a non-magnetic powder between the non-magnetic support and the magnetic layer.
 4. The magnetic tape according to claim 3, wherein the non-magnetic powder of the non-magnetic layer includes an Fe-based inorganic oxide powder having an average particle volume of 2.0×10^{-6} m³ or less.
 5. The magnetic tape according to claim 3, wherein the non-magnetic powder of the non-magnetic layer includes carbon black having a pH of 9.0 or less.
 6. The magnetic tape according to claim 3, wherein a thickness of the non-magnetic layer is 0.1 μm or more and 0.7 μm or less.
 7. The magnetic tape according to claim 1, further comprising: a back coating layer containing a non-magnetic powder on a surface side of the non-magnetic support opposite to a surface side having the magnetic layer.
 8. The magnetic tape according to claim 7, wherein the non-magnetic powder of the back coating layer includes a non-magnetic powder having an average plate diameter of 50 nm or more and 1,000 nm or less and an average plate thickness of 12 nm or less.
 9. The magnetic tape according to claim 1, wherein a tape thickness is 5.2 μm or less.
 10. The magnetic tape according to claim 1, wherein a tape thickness is 5.0 μm or less.
 11. The magnetic tape according to claim 1, wherein a vertical squareness ratio of the magnetic tape is 0.60 or more.
 12. The magnetic tape according to claim 1, wherein a vertical squareness ratio of the magnetic tape is 0.65 or more.
 13. The magnetic tape according to claim 1, wherein the standard deviation σ of the Vvc is 0.095 nm³ or less, the magnetic tape further includes a non-magnetic layer containing a non-magnetic powder between the non-magnetic support and the magnetic layer, the non-magnetic powder of the non-magnetic layer includes an Fe-based inorganic oxide powder having an average particle volume of 2.0×10^{-6} m³ or less and carbon black having a pH of 9.0 or less, a thickness of the non-magnetic layer is 0.1 μm or more and 0.7 μm or less, the magnetic tape further includes a back coating layer containing a non-magnetic powder on a surface side of the non-magnetic support opposite to a surface side having the magnetic layer, the non-magnetic powder of the back coating layer includes a non-magnetic powder having an average plate diameter of 50 nm or more and 1,000 nm or less and an average plate thickness of 12 nm or less, a tape thickness is 5.0 μm or less, and a vertical squareness ratio of the magnetic tape is 0.65 or more.
 14. A magnetic tape cartridge comprising: the magnetic tape according to claim 1.
 15. A magnetic tape apparatus comprising: the magnetic tape according to claim 1.
 16. The magnetic tape apparatus according to claim 15, further comprising: a magnetic head, wherein the magnetic head has a module including an element array with a plurality of magnetic head elements between a pair of servo signal reading elements, and the magnetic tape apparatus changes an angle θ formed by an axis of the element array with respect to a width direction of the magnetic tape during running of the magnetic tape in the magnetic tape apparatus.
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